Secant defects with prescribed holes: arcs, caps, and matching designs

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Abstract

Let A be an arc in a finite projective plane of order q and let $\mathcal{H}$ be any prescribed set of points disjoint from A. The first two secant moments give an exact defect identity: the amount by which the secant-covered points outside $A \cup \mathcal{H}$ fall short of their counting ceiling is a sum of nonnegative local terms. When $|\mathcal{H}| = q + 1$ and the secants cover everything outside $A \cup \mathcal{H}$, it yields

$$|A| \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}},$$

and the constant $3/2$ survives any $o(q^{3/2})$ exempt points. At zero defect the concurrency classes of secants are maximum matchings forming a simple $\mathrm{MATCH}(k, \lfloor k/2 \rfloor, 1)$ design, realized in Desarguesian planes by concurrent chord families; positive defect gives quantitative deletion stability, and Hirzebruch's inequality shows that no such design with at least six points is realized in characteristic zero. For caps in $\mathrm{PG}(n, q)$ the identity persists with coplanar quadruples in place of $\binom{k}{4}$, giving $|A| \geq \sqrt{2}\,q + \frac{1}{2} + \frac{3}{\sqrt{2}} - o(1)$ for complete caps in $\mathrm{PG}(3, q)$, against $\frac{1}{2} + \sqrt{2}$ from the counting bound.

For holes on a nonsingular conic $\mathcal{C} \subseteq \mathrm{PG}(2, q)$, let $\rho_{\mathcal{C}}(q)$ denote the minimum size. In characteristic two, a zero-defect $\mathcal{C}$-complete arc with odd $k \geq 7$ is an oval whose nucleus lies on $\mathcal{C}$, and even $k \geq 6$ forces a hyperoval. The coordinate triangle gives $\rho_{\mathcal{C}}(3) = 3$; a kernel-checked classification gives $\rho_{\mathcal{C}}(16) = 9$, and exhaustive classifications with kernel-checked witnesses give $\rho_{\mathcal{C}}(13) = 8$, $\rho_{\mathcal{C}}(17) = 9$, and $\rho_{\mathcal{C}}(19) = 10$. The defect identity, equality criterion, and deletion stability are formalized in Lean.

Keywords. complete arc; conic; saturating set; secant equations; matching design; oval; nucleus; finite projective plane.

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1 Introduction

An arc in a projective plane is a set of points no three of which are collinear. An arc is complete when it cannot be enlarged while remaining an arc; equivalently, every point outside it lies on a secant determined by two of its points. Complete arcs and the closely related problem of finding small saturating sets have a substantial literature; see, for example, Ball [9], Kim–Vu [16], and Nagy [20].

More generally, one may prescribe a set of points that are allowed to remain uncovered. The exact defect identity for this problem, valid in every finite projective plane and for an arbitrary prescribed set, is the main result of the paper. Its principal application here takes the prescribed set to be a nonsingular conic $\mathcal{C} \subset \mathrm{PG}(2, q)$.

Definition 1.1. An arc $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ is *$\mathcal{C}$-complete* if every point of

$$\mathrm{PG}(2,q) \setminus (A \cup \mathcal{C})$$

lies on a secant of A. Define

$$\rho_{\mathcal{C}}(q) = \min\{|A| : A \text{ is a } \mathcal{C}\text{-complete arc}\}.$$

More generally, if $\mathcal{H}$ is any set disjoint from an arc A, write

$$\mathcal{U}_{\mathcal{H}}(A) = \mathrm{PG}(2,q) \setminus \left(A \cup \mathcal{H} \cup \bigcup_{\{a,b\} \in \binom{A}{2}} ab \right).$$

We abbreviate $U(A) = \mathcal{U}_{\varnothing}(A)$ for the ordinary-uncovered locus and retain $\mathcal{U}_{\mathcal{C}}(A)$ for the conic-relative locus. Thus A is $\mathcal{C}$-complete exactly when $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$.

The minimum is attained because the finite point set $\mathrm{PG}(2,q) \setminus \mathcal{C}$ contains an inclusion-maximal arc A. For every point $x \notin A \cup \mathcal{C}$, maximality forces $A \cup \{x\}$ to contain three collinear points, so x lies on a secant of A. Thus A is $\mathcal{C}$-complete. All nonsingular conics in $\mathrm{PG}(2,q)$ are projectively equivalent, so the numerical value of $\rho_{\mathcal{C}}(q)$ does not depend on the chosen conic.

Theorem 1.2 (Exact prescribed-hole defect)**.** *Let A be a k-arc, $k \geq 3$, in a projective plane Π of order q, and let $\mathcal{H} \subseteq \Pi \setminus A$. For $x \in \Pi \setminus A$, let $r(x)$ be the number of secants of A through x, and put*

$$m = \lfloor k/2 \rfloor, \qquad N = \binom{k}{2},$$

let $\mathcal{X}_{\mathcal{H}}(A)$ be the points of $\Pi \setminus (A \cup \mathcal{H})$ lying on a secant of A, and put $I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y)$. Define

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

Then

$$m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).$$

Corollary 1.3 (Equality rigidity and deletion stability)**.** *Label the secants by the edges of K_k. Two secants with disjoint endpoint pairs are adjacent in the Kneser graph $KG(k,2)$, and their concurrency cliques canonically decompose its edges. For $k \geq 4$, if $\Delta_{\mathcal{H}}(A) = 0$, these cliques are maximum matchings and form a simple $\mathrm{MATCH}(k, \lfloor k/2 \rfloor, 1)$ design. When the plane is Desarguesian over K, the original k-arc realizes this design in $\mathrm{PG}(2,K)$: the chords indexed by every matching block are concurrent. This is a rank-three projective realization. Dually, edges sharing a vertex lie on star lines and disjoint edges lie on matching lines; this is the star–matching incidence realization. The defect controls both the edge count and the vertex-cover number of the graph formed by the nonmaximum concurrency cliques; after deleting at most $m\Delta_{\mathcal{H}}(A)$ secants, every surviving Kneser edge lies in a maximum concurrency clique.*

Theorem 1.4 ($q+1$-hole lower bound)**.** *Let Π be a projective plane of order q, let $\mathcal{H} \subseteq \Pi$ have $q+1$ points, and let $A \subseteq \Pi \setminus \mathcal{H}$ be a k-arc such that every point of $\Pi \setminus (A \cup \mathcal{H})$ lies on a secant of A. If*

$$L_2(q) = \min\left\{ j \geq 3 : q^2 - j \leq \binom{j}{2}(q-1) - \frac{6}{\lfloor j/2 \rfloor}\binom{j}{4} \right\},$$

then $k \geq L_2(q)$. In particular, every nonsingular conic satisfies

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.$$

Theorem 1.5 (Exact small orders)**.**

$$\rho_{\mathcal{C}}(3) = 3, \qquad \rho_{\mathcal{C}}(13) = 8, \qquad \rho_{\mathcal{C}}(16) = 9, \qquad \rho_{\mathcal{C}}(17) = 9, \qquad \rho_{\mathcal{C}}(19) = 10.$$

The universal identity, equality criterion, and stability statements have ordinary proofs in the text. The exact small orders have separate finite inputs: the order-16 lower exclusion is an exhaustive kernel-checked certificate, whereas the lower exclusions at orders $13, 17, 19$ are trusted classifier executions with independently checked attaining witnesses. Appendix F records these two trust boundaries once.

Proof map. The argument has four moves. The first two classical secant moments are split between required points and allowed holes; one linear combination then factors pointwise into nonnegative terms. Vanishing turns the concurrence cliques of the secants into maximum matchings, while the same summands bound the edges and secants that obstruct this pattern away from equality. Finally, specializing the holes to a conic gives the scalar bound and, when the remainder vanishes, parity-dependent order equations; the difficult characteristic-two branch closes by a chord-involution sign argument. Thus the identity extracts equality and stability from the classical moments; it is not an independent third incidence equation.

The same machinery reaches further than the conic. The defect splits into an integer intrinsic part and a hole part, which sharpens the discrete gap (Proposition 3.6); the constant $3/2$ persists for any $o(q^{3/2})$ prescribed holes and degrades linearly beyond that scale (Theorem 4.10); the equality designs are never realized in characteristic zero once $k \geq 6$, by Hirzebruch's inequality for line arrangements (Theorem A.5), and the ten-point design has a projectively unique realization (Corollary A.4). In $\mathrm{PG}(n, q)$ the identity holds for caps with the number of coplanar quadruples in place of $\binom{k}{4}$, and counting those quadruples secant by secant yields a coverage bound with a term the plane does not have; in $\mathrm{PG}(3, q)$ it raises the additive constant of the counting bound for complete caps from $\frac{1}{2} + \sqrt{2}$ to $\frac{1}{2} + \frac{3}{\sqrt{2}}$ (Section 5).

Optional coding translation. For $k \geq 4$, projective representatives of the points of A in a Desarguesian plane form a parity-check matrix of a $[k, k-3, 4]_q$ MDS code. If $U(A) \neq \varnothing$, then $U(A)$ is its projective deep-hole syndrome locus: points of A, points outside A on secants, and points of $U(A)$ have coset weight one, two, and three, respectively. Quantitatively, a syndrome whose projective point x lies outside A is realized by exactly $r(x)$ error vectors of weight two, one on each secant through x; and when A is $\mathcal{C}$-complete at most $(q^2-1)/(q^3-1)$ of the nonzero syndromes need weight three, since their points lie on $\mathcal{C}$. This is a dictionary for the uncovered-locus statements, not an input to their proofs.

The surrounding notions differ at the defining containment. Saturating sets need not be arcs, while almost-complete conic subsets select points on the conic [11]. A line hole recovers affine completeness; see Corollary 3.12. Hyperfocusedness is a different secant-blocking condition, requiring a minimum focus set on an external line. For translation arcs, Giulietti and Montanucci separately relate hyperfocusedness to affine extendability [13, Definition 2 and Proposition 3.4]; see also [21, 17]. Complete exterior sets point in the opposite direction: their joins miss the conic [12, p. 143]. Thus complete exteriority gives $\mathcal{C}(\mathbb{F}_q) \subseteq U(A)$, whereas $\mathcal{C}$-completeness is equivalent to $U(A) \subseteq \mathcal{C}(\mathbb{F}_q)$.

The point-index equations themselves are classical; compare Hirschfeld [15, Chapter 9], Ball [9, p. 34], and the recent explicit complete-arc bound of Alabdullah–Hirschfeld [1]. The new step is the exact local remainder after splitting those equations over a prescribed hole and its complement.

Localization of an uncovered locus on a proper subgeometry is also known for curve-derived arcs with larger line intersections. Korchmáros, Nagy, and Szőnyi [18, Theorem 7.5] prove that, for odd q and odd $r \geq 5$, the points left uncovered by the $(q+1)$-secants of their rational BKS $(k, q+1)$-arc in $\mathrm{PG}(2, q^r)$ are exactly the points of the proper subplane $\mathrm{PG}(2, q)$ not belonging to the arc. Thus localization on the defining subfield subplane is prior art. Their theorem concerns a particular curve-derived $(k, q+1)$-arc, not an arbitrary-hole remainder for ordinary arcs.

The equality structure belongs to the established theory of matching designs. Alspach and Heinrich [3] call a family of s-edge matchings in K_n a $\mathrm{MATCH}(n, s, \lambda)$ design when every pair of independent edges occurs in exactly λ members. Their framework generalizes hyperfactorizations, which were introduced in connection with higher block designs. Here the prescribed-hole equality condition selects matching blocks realized by secant concurrences in one projective plane. After duality, Corollary 3.8 makes this the full star–matching pairwise-balanced design and, in a Desarguesian plane, a rank-three realization over the underlying field. The resulting field and characteristic restrictions are recorded in Appendix A.

For a first pass, read Sections 2–4 and the conclusion: they contain the complete route from exact accounting through equality rigidity and stability to the conic bound and equality spectra. A reader interested only in that mechanism may stop there. The exact finite values continue in Appendix D; the equality-realization classification, upper-bound transfer, nucleus refinements, inverse reconstruction, verification contracts, and coordinate witnesses are independent secondary layers and may be read separately.

2 The classical secant equations

Throughout the remainder of the paper, $q \geq 3$ and $k \geq 3$. Let Π be a projective plane of order q, and let A be a k-arc in Π. Write $r_A(x)$ for the number of secants of A through $x \in \Pi \setminus A$. We usually write $r(x)$. Put

$$N = \binom{k}{2}, \qquad m = \left\lfloor \frac{k}{2} \right\rfloor.$$

Lemma 2.1. *For every $x \notin A$, one has $r(x) \leq m$.*

Proof. Distinct secants through x cannot share an endpoint in A. Indeed, two lines through the same pair x, a coincide. Hence the secants through x use pairwise disjoint pairs of points of A, of which there are at most $\lfloor k/2 \rfloor$. □

Proposition 2.2 (Classical first and second index equations)**.** *For every k-arc A in a projective plane of order q,*

$$\sum_{x \notin A} r(x) = \binom{k}{2}(q-1), \tag{1}$$

$$\sum_{x \notin A} \binom{r(x)}{2} = 3\binom{k}{4}. \tag{2}$$

These equations are the standard first two secant-index moments; see Hirschfeld [15, Chapter 9] and Ball [9, p. 34]. We include the proof to fix the indexing convention used in the prescribed-hole split.

Proof. There are $\binom{k}{2}$ secants. Each contains $q+1$ points, exactly two of them in A, and therefore contributes $q-1$ incidences with points outside A. This proves (1).

For (2), count unordered pairs of distinct secants through a common external point. By Lemma 2.1, their four endpoints in A are distinct. Conversely, a four-subset of A has three partitions into two unordered pairs. Each partition gives two secants, which meet in a unique point because the plane is projective. Their intersection is outside A: otherwise one of the two secants would contain three points of the arc. Thus every four-subset contributes exactly three. $\square$

Only the projective-plane axioms and the common line size $q+1$ enter these proofs. In particular, the second equation need not hold in a general finite linear space where two lines may be disjoint.

3 An exact defect identity for prescribed holes

The next theorem is stated for an arbitrary exceptional point set. The conic enters only in later specializations.

Let $\mathcal{H} \subseteq \Pi \setminus A$ be any set of points allowed to remain uncovered. The notation used in the identity is summarized by

$$\begin{array}{c|l} V_{\mathcal{H}}(A) = \Pi \setminus (A \cup \mathcal{H}) & \text{required locus} \\ \mathcal{X}_{\mathcal{H}}(A) = \{x \in V_{\mathcal{H}}(A) : r(x) > 0\} & \text{covered required points} \\ \mathcal{U}_{\mathcal{H}}(A) = V_{\mathcal{H}}(A) \setminus \mathcal{X}_{\mathcal{H}}(A) & \text{uncovered required points} \\ I_{\mathcal{H}}(A) = \sum_{y \in \mathcal{H}} r(y) & \text{secant incidence spent on the holes.} \end{array}$$

The third line agrees with the uncovered-locus notation introduced in Definition 1.1; in particular, $U = \mathcal{U}_{\varnothing}$ and the conic-relative locus is $\mathcal{U}_{\mathcal{C}}$.

Splitting Proposition 2.2 over $V_{\mathcal{H}}(A)$ and $\mathcal{H}$ gives

$$\sum_{x \in V_{\mathcal{H}}(A)} r(x) = N(q-1) - I_{\mathcal{H}}(A), \tag{3}$$

$$\sum_{x \in V_{\mathcal{H}}(A)} \binom{r(x)}{2} + \sum_{y \in \mathcal{H}} \binom{r(y)}{2} = 3\binom{k}{4}. \tag{4}$$

Theorem 3.1 (Prescribed-hole defect identity). *Let A be a k-arc in a projective plane of order q, let $\mathcal{H} \subseteq \Pi \setminus A$, and put $m = \lfloor k/2 \rfloor$. Define*

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|.$$

It is the exact coverage capacity left after the universal second-moment and hole-incidence losses; the identity below resolves that capacity into local nonextremal-index contributions. Then

$$\boxed{m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m-r(y)).} \tag{5}$$

In particular, $\Delta_{\mathcal{H}}(A) \geq 0$.

Proof. Only points of $\mathcal{X}_{\mathcal{H}}(A)$ contribute to the sum in (3). Multiplying the definition of $\Delta_{\mathcal{H}}(A)$ by m and using (3) gives

$$m\Delta_{\mathcal{H}}(A) = m \sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} r(x) + (m-1)I_{\mathcal{H}}(A) - m|\mathcal{X}_{\mathcal{H}}(A)| - 6\binom{k}{4}.$$

By (4),

$$6\binom{k}{4} = 2\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \binom{r(x)}{2} + 2\sum_{y\in\mathcal{H}} \binom{r(y)}{2}.$$

Substitution first separates the required-point and hole terms:

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) = &\sum_{x\in\mathcal{X}_{\mathcal{H}}(A)} \left(m(r(x)-1) - 2\binom{r(x)}{2}\right) \\ &+ \sum_{y\in\mathcal{H}} \left((m-1)r(y) - 2\binom{r(y)}{2}\right). \end{aligned}$$

The pointwise identities

$$\begin{aligned} m(r-1) - 2\binom{r}{2} &= (r-1)(m-r), \\ (m-1)r - 2\binom{r}{2} &= r(m-r), \end{aligned}$$

now give (5). Every term is nonnegative by Lemma 2.1. □

For required points, the identity records the total slack in the pointwise quadratic inequality

$$\mathbf{1}_{\{r>0\}} \le r - \frac{2}{m}\binom{r}{2} \qquad (0 \le r \le m),$$

whose positive-index equality cases are exactly $r = 1$ and $r = m$. Thus the factored remainder is a sharp two-moment certificate, rather than only a rearrangement of the global counts.

Corollary 3.2 (Coverage and uncovered-locus bounds)**.** *Under the hypotheses of Theorem 3.1,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}, \tag{6}$$

and

$$|\mathcal{U}_{\mathcal{H}}(A)| \ge |V_{\mathcal{H}}(A)| - N(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{H}}(A)}{m}. \tag{7}$$

Equality holds in either inequality if and only if

$$r(x) \in \{1, m\} \quad (x \in \mathcal{X}_{\mathcal{H}}(A)), \qquad r(y) \in \{0, m\} \quad (y \in \mathcal{H}).$$

Proof. The two inequalities are respectively $\Delta_{\mathcal{H}}(A) \ge 0$ and its rearrangement using $|\mathcal{U}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| - |\mathcal{X}_{\mathcal{H}}(A)|$. The equality statement follows from the vanishing of every nonnegative term in (5). □

For later use, we record the complete-outside consequence before imposing any geometry on the holes.

Corollary 3.3 (Arbitrary prescribed holes)**.** *Let $|\mathcal{H}| = h$. If A is a k-arc disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 + q + 1 - k - h \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{H}}(A)}{m}. \tag{8}$$

Proof. Completeness gives $|\mathcal{X}_{\mathcal{H}}(A)| = |V_{\mathcal{H}}(A)| = q^2+q+1-k-h$. Substitute this equality into (6). □

Corollary 3.4 (Any $q+1$ prescribed holes)**.** *Let $|\mathcal{H}| = q+1$. If a k-arc A is disjoint from $\mathcal{H}$ and complete outside $\mathcal{H}$, then*

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4}.$$

Consequently $k \geq L_2(q)$. No geometric property of $\mathcal{H}$ is used.

Proof. Corollary 3.3 gives the stronger inequality with the additional nonpositive term $-I_{\mathcal{H}}(A)/m$. Drop that term. □

The exact remainder also quantifies near equality.

Corollary 3.5 (Quantitative stability)**.** *Assume $m \geq 3$ and put*

$$M = \{x \in \mathcal{X}_{\mathcal{H}}(A) : 2 \leq r(x) \leq m-1\},$$
$$J = \{y \in \mathcal{H} : 1 \leq r(y) \leq m-1\}.$$

Then

$$(m-2)|M| + (m-1)|J| \leq m\Delta_{\mathcal{H}}(A). \tag{9}$$

Proof. For $2 \leq r \leq m-1$, $(r-1)(m-r) \geq m-2$. For $1 \leq r \leq m-1$, $r(m-r) \geq m-1$. Apply these estimates to (5). □

Thus small defect forces almost every multiply covered required point to have the maximum possible multiplicity m, while almost every hole point met by a secant also has multiplicity m.

The defect splits into a part that depends only on the secant arrangement and a part paid for the prescribed holes. Write

$$D(A) = \Delta_{\varnothing}(A) = N(q-1) - \frac{6}{m}\binom{k}{4} - |\mathcal{X}_{\varnothing}(A)|$$

for the *intrinsic defect*, the defect with no holes at all.

Proposition 3.6 (Intrinsic defect and the discrete gap)**.** *Let A be a k-arc in a projective plane of order q with $k \geq 4$, and let $\mathcal{H} \subseteq \Pi \setminus A$. Then $D(A)$ is a nonnegative integer,*

$$\Delta_{\mathcal{H}}(A) = D(A) + \sum_{\substack{y\in\mathcal{H} \\ r(y)>0}} \left(1 - \frac{r(y)}{m}\right), \tag{10}$$

and $\Delta_{\mathcal{H}}(A)$ is nondecreasing in $\mathcal{H}$. Consequently

$$m\Delta_{\mathcal{H}}(A) = 0 \quad \text{or} \quad m\Delta_{\mathcal{H}}(A) \geq m-1, \tag{11}$$

and the second alternative splits further: if $D(A) \geq 1$ then $m\Delta_{\mathcal{H}}(A) \geq m$, while if $D(A) = 0$ then $m\Delta_{\mathcal{H}}(A)$ is $m-1$ times the number of holes of index one.

Proof. The quantity $\frac{6}{m}\binom{k}{4}$ equals $(k-1)(k-2)(k-3)/2$ for even k and $k(k-2)(k-3)/2$ for odd k; both are integers, so $D(A)$ is an integer, and it is nonnegative by Theorem 3.1 with $\mathcal{H} = \varnothing$. Since $\mathcal{X}_{\mathcal{H}}(A)$ is $\mathcal{X}_{\varnothing}(A)$ with the covered holes removed, $|\mathcal{X}_{\mathcal{H}}(A)| = |\mathcal{X}_{\varnothing}(A)| - \#\{y \in \mathcal{H} : r(y) > 0\}$, and substituting this into the definition of $\Delta_{\mathcal{H}}(A)$ gives (10). Every summand there is nonnegative by Lemma 2.1, which gives monotonicity.

For the gap, multiply (10) by m. If $D(A) \geq 1$ the first term alone is at least m. If $D(A) = 0$, then (5) with $\mathcal{H} = \varnothing$ forces $r(x) \in \{1, m\}$ at every covered point $x \notin A$, holes included; so each nonzero summand in $\sum_y (m - r(y))$ equals $m - 1$. $\square$

The gap (11) is sharp: a complete quadrilateral ($k = 4$, $m = 2$) has $D(A) = 0$, and a single index-one hole gives $m\Delta_{\mathcal{H}}(A) = 1$. The two alternatives have different sources. A secant arrangement that is not itself extremal costs at least one full unit of defect, whereas an extremal arrangement acquires defect only through holes met by exactly one secant, at $(m-1)/m$ each. This is the precise stability content behind the equality heuristic.

3.1 Concurrency designs at equality

The equality criterion has a global incidence interpretation. Label the vertices of K_k by the points of A, so that its edges label the secants of A. Two such edges are adjacent in the Kneser graph $KG(k, 2)$ exactly when the corresponding secants have disjoint endpoint pairs.

Theorem 3.7 (Concurrency decomposition and equality rigidity)**.** *Suppose $k \geq 4$. For each $x \notin A$ with $r(x) \geq 2$, the secants through x form an $r(x)$-edge matching M_x in K_k, and*

$$E\big(KG(k,2)\big) = \dot{\bigcup_{x \notin A,\ r(x) \geq 2}} E(K_{M_x}).$$

Here K_{M_x} is the clique whose vertices are the edges of M_x.

If $\Delta_{\mathcal{H}}(A) = 0$, every M_x in this decomposition has $m = \lfloor k/2 \rfloor$ edges. The family is simple: a repeated matching would repeat each of its pairs of edges, contradicting $\lambda = 1$. Thus $(M_x)_{r(x)=m}$ is a simple MATCH$(k, m, 1)$ *design. If $Z = \{x \notin A : r(x) = m\}$, then*

$$|Z| = \frac{3\binom{k}{4}}{\binom{m}{2}} = \begin{cases} (k-1)(k-3), & k \text{ even}, \\ k(k-2), & k \text{ odd}, \end{cases}$$

and every secant of A contains $k-3$ points of Z when k is even and $k-2$ points of Z when k is odd.

Proof. Secants through a point outside an arc have pairwise disjoint endpoint pairs, so they form a matching. Conversely, two secants with disjoint endpoint pairs meet in a unique point outside A. Their edge in $KG(k, 2)$ therefore belongs to exactly one concurrence clique, proving the decomposition.

If the defect vanishes, Theorem 3.1 gives $r(x) \in \{1, m\}$ off $\mathcal{H}$ and $r(x) \in \{0, m\}$ on $\mathcal{H}$. Every concurrence point has index at least two, so all the cliques above have order m. The second index equation now gives $|Z|\binom{m}{2} = 3\binom{k}{4}$.

Fix a secant. It is disjoint from $\binom{k-2}{2}$ other secants, and each point of Z on it accounts for $m-1$ of them. Hence it contains $\binom{k-2}{2}/(m-1)$ points of Z, which has the two displayed values. $\square$

Corollary 3.8 (Dual star–matching incidence realization)**.** *Suppose $k \geq 4$. If $\Delta_{\mathcal{H}}(A) = 0$, then after any projective duality the $\binom{k}{2}$ secants become points carrying an incidence* *realization of the star–matching pairwise-balanced design on $E(K_k)$. Its lines are the k star lines of size $k-1$, one for each point of A, and the $|Z|$ matching lines of size $m = \lfloor k/2 \rfloor$. If the original plane is Desarguesian over K, this is a rank-three projective realization over K.*

Proof. The dual secant-points for two edges of K_k lie on a star line when the edges share a vertex. When the edges are disjoint, their secants meet at a unique point outside A; Theorem 3.7 places them on a unique matching line, of size m at zero defect. □

The same decomposition gives two distances from equality. Let G_{bad} be the spanning subgraph of $KG(k,2)$ whose edges lie in concurrence cliques of order $2 \le r(x) \le m-1$, and write $\tau(G_{\mathrm{bad}})$ for its vertex-cover number. Thus its vertices are the secants of A, and a vertex cover is a set of secants meeting every Kneser edge in a nonmaximum concurrence clique.

Corollary 3.9 (Edge and vertex stability)**.** *For $k \ge 4$,*

$$|E(G_{\mathrm{bad}})| \le \frac{m(m-1)}{2}\,\Delta_{\mathcal{H}}(A), \qquad \tau(G_{\mathrm{bad}}) \le m\Delta_{\mathcal{H}}(A).$$

The second bound says equivalently that there is a set $D \subseteq \binom{A}{2}$, with $|D| \le m\Delta_{\mathcal{H}}(A)$, such that every two secants outside D with disjoint endpoint pairs meet at a point of secant index m.

Proof. At an off-hole centre of index r,

$$\binom{r}{2} \le \frac{m-1}{2}(r-1)(m-r).$$

For a hole centre whose clique contributes to G_{bad}, its defect weight $r(m-r)$ is at least the off-hole weight $(r-1)(m-r)$. Summing over the edge-disjoint concurrence cliques and applying (5) proves the edge bound.

At each concurrence centre x with $2 \le r(x) \le m-1$, retain one secant through x and put the other $r(x)-1$ secants into D. This is a vertex cover of G_{bad}. For an off-hole centre its cost satisfies

$$r(x) - 1 \le (r(x)-1)(m - r(x)),$$

and for a hole centre it satisfies

$$r(x) - 1 \le r(x)(m - r(x)).$$

Summing and applying (5) gives $|D| \le m\Delta_{\mathcal{H}}(A)$. Any two surviving secants with disjoint endpoint pairs meet at a concurrence centre of index at least two. That centre cannot have index below m, since all but one of its secants would then meet D; hence its index is m. □

Corollary 3.10 (Matching-design nonexistence gives a two-unit gap)**.** *Let $k \ge 4$, put $m = \lfloor k/2 \rfloor$, and suppose that*

$$v = \frac{3\binom{k}{4}}{\binom{m}{2}}$$

is an integer. If no simple MATCH$(k,m,1)$ *design exists, then every k-arc and every prescribed hole set disjoint from it satisfy*

$$\Delta_{\mathcal{H}}(A) \ge 2.$$

Proof. The maximum-index concurrence cliques form a packing of b maximum matchings in $KG(k,2)$. If B is the number of Kneser edges in the remaining concurrence cliques, the exact concurrence decomposition gives

$$B = (v-b)\binom{m}{2}.$$

Corollary 3.9 gives $B \le \binom{m}{2}\Delta_{\mathcal{H}}(A)$, so $v - b \le \Delta_{\mathcal{H}}(A)$.

It remains to exclude $v - b = 1$. The leave of such a packing has $\binom{m}{2}$ edges. Every positive leave degree is divisible by $m-1$: the ambient Kneser degree and the degree removed by each K_m block are both divisible by $m-1$. Hence the leave has at most m nonisolated vertices, while its minimum positive degree forces at least m. It is therefore a K_m, supported on m pairwise disjoint edges of K_k, and adjoining that maximum matching completes the design. Nonexistence thus forces $v-b \ge 2$, proving the claim. □

Remark 3.11 (Slack in the deletion bound). The proof records why the vertex-cover bound can be strict. Include hole points of index 1 with zero deletion cost, let $\mathcal{B}_R$ be the off-hole centres with $2 \le r < m$, and let $\mathcal{B}_H$ be the hole points with $1 \le r < m$. If D is the cover constructed above and

$$\Omega = \sum_{z \in \mathcal{B}_R \cup \mathcal{B}_H} (r(z) - 1) - |D|,$$

then $\Omega \ge 0$ measures overlap among the local deletion sets, and

$$\begin{aligned} m\Delta_{\mathcal{H}}(A) - |D| = \Omega &+ \sum_{x \in \mathcal{B}_R} (r(x)-1)(m - r(x) - 1) \\ &+ \sum_{y \in \mathcal{B}_H} \big(r(y)(m - r(y)) - (r(y) - 1)\big). \end{aligned}$$

Thus equality for this cover requires no intermediate-index hole point, requires every bad off-hole centre to have index $m-1$, and requires the local deletion sets to be disjoint. The identity does not assert that these necessary local conditions are globally realizable.

The line-hole case now makes the connection with complete affine arcs explicit.

Corollary 3.12 (Complete affine arcs)**.** *Let L_∞ be a line in a projective plane of order q, and let A be a k-arc disjoint from L_∞. Then A is complete as an arc in the affine plane $\Pi \setminus L_\infty$ if and only if it is complete outside L_∞. In that case*

$$q^2 - k \le \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}\binom{k}{2}. \tag{12}$$

Equality holds if and only if every affine point outside A has secant index in $\{1, m\}$ and every point of L_∞ has secant index in $\{0, m\}$. If moreover $k \ge 6$ is even and equality holds, then

$$q \in \left\{ k-2, \binom{k-1}{2} + 1 \right\}.$$

Proof. Maximality in the affine plane is equivalent to every affine point outside A lying on a secant of A, which is exactly completeness outside L_∞. Every secant meets L_∞ once, so $I_{L_\infty}(A) = \binom{k}{2}$. Now apply Corollaries 3.3 and 3.2; the latter also gives the equality criterion. For even k, substituting the fixed value of $I_{L_\infty}(A)$ into the zero-defect equation and factoring gives

$$(q - (k-2)) \left(q - \left(\binom{k-1}{2} + 1 \right) \right) = 0.$$

□

4 Specialization to a conic

Let $\mathcal{C}$ be a nonsingular conic in $\mathrm{PG}(2,q)$, and let $A \subseteq \mathrm{PG}(2,q) \setminus \mathcal{C}$ be a k-arc. Since $|\mathcal{C}| = q+1$, there are $q^2 - k$ points outside $A \cup \mathcal{C}$. Write

$$I_{\mathcal{C}}(A) = \sum_{y \in \mathcal{C}} r(y) = \sum_{\ell \text{ an } A\text{-secant}} |\ell \cap \mathcal{C}|.$$

The specialization has two layers. Dropping the nonnegative conic-incidence term gives the universal $q+1$-hole lower bound; retaining it at zero defect forces the parity-dependent order spectra. Conic geometry is needed only for the equality classification and finite applications, not for the defect identity or its scalar correction.

Corollary 4.1 (Conic-loss and uncovered-locus inequalities)**.** *For every such arc,*

$$|\mathcal{U}_{\mathcal{C}}(A)| \geq q^2 - k - \binom{k}{2}(q-1) + \frac{6}{m}\binom{k}{4} + \frac{I_{\mathcal{C}}(A)}{m}. \tag{13}$$

If A is $\mathcal{C}$-complete, then

$$\boxed{q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{I_{\mathcal{C}}(A)}{m}.} \tag{14}$$

Proof. Apply Corollary 3.2 with $\mathcal{H} = \mathcal{C}$, using $|V_{\mathcal{C}}(A)| = q^2 - k$ and $I_{\mathcal{H}}(A) = I_{\mathcal{C}}(A)$. If A is $\mathcal{C}$-complete, then $\mathcal{U}_{\mathcal{C}}(A) = \varnothing$, and the first inequality rearranges to (14). □

The first correction in (14) accounts for unavoidable concurrence among secants; the last accounts for secant incidence spent on the exempt conic. Dropping the last term recovers Corollary 3.4.

Corollary 4.2 (Even-size equality spectrum)**.** *Let $k \geq 6$ be even. If a $\mathcal{C}$-complete k-arc has zero defect, then*

$$q \in \left\{ k-2,\ \binom{k-1}{2},\ \binom{k-1}{2} + 1 \right\}.$$

In the first two cases every point of $\mathcal{C}$ has index $k/2$; in the third, exactly $k-1$ conic points have that index.

Proof. Put $Q = k-2$ and let $Z = \{x \notin A : r(x) = m\}$. Then $m = Q/2 + 1$ and, by Theorem 3.7, $|Z| = Q^2 - 1$. Put $s = |Z \cap \mathcal{C}|$. Every positive conic index equals m, so $I_{\mathcal{C}}(A) = ms$. Substitution in the zero-defect identity and factorization give

$$s = q + 1 - \frac{(q-Q)(2q - Q(Q+1))}{2}.$$

The elementary arc bound $k \leq q+2$ gives $q \geq Q$. Since $0 \leq s \leq q+1$, either $q = Q$ or $q \geq B := Q(Q+1)/2$. Write $q = B + t$ in the latter case. Then

$$s = q + 1 - t(q-Q).$$

For $t \geq 2$, already at $t = 2$ the subtracted term exceeds $q+1$ by

$$B - 2Q + 1 = \frac{(Q-1)(Q-2)}{2} > 0,$$

and the excess increases with t. Hence $t \in \{0,1\}$. The displayed values of s at $q = Q, B, B+1$ are respectively $q+1, q+1, Q+1$. □

Corollary 4.3 (Odd-size equality spectrum)**.** *Let $k \geq 7$ be odd. If a $\mathcal{C}$-complete k-arc has zero defect, then*

$$q \in \left\{ k-1,\ \binom{k-1}{2} - 1,\ \binom{k-1}{2} \right\}.$$

If s is the number of conic points of index $(k-1)/2$, then the corresponding values of s are respectively $q, q, k-1$.

Proof. Put $Q = k-1$, $m = Q/2$, and $R = \binom{Q}{2} - 1$. At zero defect every positive conic index equals m, and substitution in the defect identity gives

$$s = q - (q-Q)(q-R).$$

The arc bound gives $q \geq Q-1$, but the right-hand side is negative at $q = Q-1$, so $q \geq Q$. If $Q \leq q \leq R$, then $s \leq q+1$ gives $(q-Q)(R-q) \leq 1$. Since $R - Q > 2$, this forces $q = Q$ or $q = R$. If $q = R + t$ with $t \geq 1$, then $s = q - t(q-Q)$. The value $t = 1$ is possible, whereas $t \geq 2$ and $s \geq 0$ would imply $q \leq 2Q$, contrary to $q \geq R + 2 > 2Q$. Substitution gives the stated values of s. □

Corollary 4.4 (Odd equality in characteristic two)**.** *Let q be a power of two, let $k \geq 7$ be odd, and let A be a k-arc disjoint from $\mathcal{C}$. Then A is $\mathcal{C}$-complete with zero defect if and only if $k = q+1$, A is an oval, and its nucleus belongs to $\mathcal{C}$.*

Proof. Put $Q = k-1$. The value $\binom{Q}{2}$ in Corollary 4.3 is not a power of two: the coprime factors $Q/2$ and $Q-1 > 1$ would both have to be powers of two. Nor can $2^n = \binom{Q}{2} - 1$. Writing $Q = 2a$ and $x = 4a - 1$ would give

$$(x-3)(x+3) = 2^{n+3};$$

the two positive powers of two differ by six, hence are 2 and 8, which gives $x = 5$, incompatible with integral a. Thus $q = Q$.

A $(q+1)$-arc in even order is an oval. Its nucleus is the unique external point of secant index zero, and every other external point has index $q/2$. The value $s = q$ from Corollary 4.3 therefore says exactly that the prescribed conic contains the nucleus. Conversely, an oval disjoint from $\mathcal{C}$ whose nucleus lies on $\mathcal{C}$ has the required index pattern, so it is $\mathcal{C}$-complete with zero defect. □

Corollary 4.5 (Characteristic-two equality spectrum)**.** *Let $k \geq 6$ be even, let q be a power of two, and suppose that a $\mathcal{C}$-complete k-arc has zero defect. Then $k = q+2$.*

The proof first removes one candidate order arithmetically. In the remaining exceptional order, parity excludes $\nu \in A$ and then forces $r(\nu) = m$, hence a secant of A tangent to the conic. The induced chord involutions preserve the maximum-index conic points, and their permutation signs are incompatible.

Proof. Put $Q = k-2$ and $B = Q(Q+1)/2$. By Corollary 4.2, $q \in \{Q, B, B+1\}$. Write $q = 2^n$. The case $q = B$ is impossible: since Q is even, the coprime factors Q and $Q+1$ could have product a power of two only if $Q + 1 = 1$.

Suppose $q = B+1$, and write $k = 2m$. Then

$$q = (m-1)(2m-1) + 1.$$

Since $q > 1$ is even, m is even. The last clause of Corollary 4.2 gives $s = k - 1$ conic points of index m, and hence

$$I_{\mathcal{C}}(A) = ms = \binom{k}{2}.$$

Let ν be the nucleus of $\mathcal{C}$. If $\nu \in A$, then the $k - 1$ lines from ν to the other arc points are exactly the conic-tangent secants of A: any further conic-tangent secant would contain ν and two other arc points. All conic-nontangent secants meet $\mathcal{C}$ in zero or two points, so $I_{\mathcal{C}}(A) \equiv k - 1 \pmod 2$, impossible because m is even. Thus $\nu \notin A$. Now the conic-tangent secants of A are exactly the secants through ν, and therefore $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. Zero defect gives $r(\nu) \in \{1, m\}$, so $r(\nu) = m$. Hence an arc secant PQ is tangent to $\mathcal{C}$, say at T.

Projection from a point $R \notin \mathcal{C}$ pairs the two intersections with $\mathcal{C}$ of every chord through R; a tangent contact is paired with itself. This defines an involution σ_R of $\mathcal{C}(\mathbb{F}_q)$. In characteristic two, σ_ν is the identity, whereas for $R \neq \nu$ the unique conic tangent through R gives the unique fixed point of σ_R.

Let S be the set of maximum-index conic points. This set is invariant under every σ_R with $R \in A$. Indeed, if $Y \in S$, the secants through Y form a perfect matching of A, so R has a partner $R' \in A$. The second conic intersection of RR', with a tangent contact counted as fixed, is $\sigma_R(Y)$; it lies on an A-secant, and zero defect forces its positive conic index to be m. The tangent contact T itself lies on the secant PQ, so it has positive conic index; zero defect therefore gives $T \in S$.

The involutions σ_P and σ_Q fix T. A projective change of coordinates makes their product law explicit. Move $\mathcal{C}$ to the standard conic and T to $(1, 0, 0)$ on $XZ = Y^2$, and write the other conic points as $c(u) = (u^2, u, 1)$. The tangent at T is $Z = 0$, and the chord through $c(u)$ and $c(v)$ meets it at $(u+v, 1, 0)$. Thus a centre $R = (a, 1, 0)$ induces $\sigma_R : u \mapsto u+a$. The value $a = 0$ is the nucleus and gives the identity; since $P, Q \in A$, the preceding exclusion of the nucleus and $P \neq Q$ give distinct nonzero parameters a_P, a_Q. Hence σ_P, σ_Q, and their product are the three nonidentity translations with parameters $a_P, a_Q, a_P + a_Q$. They commute and each has T as its unique fixed point. All three preserve S, and

$$|S| = k - 1 \equiv 3 \pmod 4.$$

Restricted to S, each nonidentity involution has one fixed point and therefore an odd number of transpositions. Thus all three permutations have sign -1, contradicting $\operatorname{sgn}(\sigma_P\sigma_Q) = \operatorname{sgn}(\sigma_P)\operatorname{sgn}(\sigma_Q) = 1$. The case $q = B + 1$ is therefore impossible. The remaining case $q = Q$ is $k = q + 2$. □

In the first branch, A is a hyperoval. Every point outside A has secant index $(q + 2)/2$, so every nonsingular conic disjoint from A produces a $\mathcal{C}$-complete zero-defect pair. Thus the oval and hyperoval branches in characteristic two are completely geometric.

Corollary 4.6 (Small even equality cases)**.** *No zero-defect $\mathcal{C}$-complete eight- or twelve-arc exists in a Desarguesian plane. A zero-defect ten-arc can occur only in* $\mathrm{PG}(2, 8)$*; its underlying arc has the regular-hyperoval concurrence design, and such equality pairs exist. This statement does not classify either their projective realizations or the prescribed conics disjoint from them.*

Moreover, if $Q \geq 4$ is an even prime power and $d \geq 2$, no $\mathcal{C}$-complete $(Q + 2)$-arc in $\mathrm{PG}(2, Q^d) \setminus \mathcal{C}$ *has zero defect.*

Proof. For $k = 8, 10, 12$, the three candidate orders are respectively

$$\{6, 21, 22\}, \qquad \{8, 36, 37\}, \qquad \{10, 55, 56\}.$$

Since a Desarguesian plane has prime-power order, this excludes $k = 8, 12$ and leaves only $q = 8, 37$ when $k = 10$.

For the final assertion,

$$Q^d \geq Q^2 > \frac{Q(Q+1)}{2} + 1$$

when $Q \geq 4$ and $d \geq 2$, so none of the three orders allowed by Corollary 4.2 equals Q^d.

For the ten-point refinement, Theorem 3.7 produces a rank-three MATCH$(10, 5, 1)$ design. Apply Theorem A.3.

For existence over $\mathbb{F}_8 = \mathbb{F}_2(\alpha)$, where $\alpha^3 + \alpha + 1 = 0$, distinguish the constituent conic

$$\mathcal{D}: \; XZ = Y^2$$

from the prescribed conic, and denote its nucleus by $N_{\mathcal{D}} = (0, 1, 0)$. Let

$$A = \mathcal{D} \cup \{N_{\mathcal{D}}\}, \qquad \mathcal{C}: \; X^2 + Y^2 + Z^2 + (1+\alpha)YZ = 0.$$

The gradient of the latter quadratic is $(0, (1+\alpha)Z, (1+\alpha)Y)$, so $\mathcal{C}$ is nonsingular. Its values on the eight affine points $(t^2, t, 1)$ of A, in the polynomial-basis order $t = 0, 1, \alpha, 1 + \alpha, \ldots, 1 + \alpha + \alpha^2$, are

$$1,\; \alpha,\; 1+\alpha^2,\; \alpha+\alpha^2,\; \alpha,\; 1,\; \alpha+\alpha^2,\; 1+\alpha^2,$$

and its value at each of the two infinite points of A is 1. Hence $A \cap \mathcal{C} = \varnothing$. Every point outside a hyperoval in PG$(2, 8)$ lies on five of its secants: the ten hyperoval points pair on the secants through that point. Thus every point of $\mathcal{C}$ has index five, so A is $\mathcal{C}$-complete and has zero defect. □

Corollary 4.7 (Explicit integer-valued lower bound)**.** *Define*

$$L_1(q) = \min\left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1) \right\},$$
$$L_2(q) = \min\left\{ k \geq 3 : q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} \right\}.$$

Then

$$\rho_{\mathcal{C}}(q) \geq L_2(q) \geq L_1(q). \tag{15}$$

Proof. For $q \geq 3$ a set of at most two points has at most one secant, and that secant together with the set and the conic contains at most $2q + 2 < q^2 + q + 1$ points. Thus a $\mathcal{C}$-complete arc has at least three points, so the moment bound applies to the unrestricted minimum defining $\rho_{\mathcal{C}}(q)$. The definition of $L_2(q)$ involves only integer arithmetic after multiplication by $\lfloor k/2 \rfloor$. The correction subtracted in the definition of $L_2(q)$ is nonnegative, so every integer feasible for $L_2(q)$ is feasible for $L_1(q)$; hence $L_2(q) \geq L_1(q)$. □

For calculation, the corrected capacity has the parity-dependent forms

$$\binom{k}{2}(q-1) - \frac{6}{\lfloor k/2 \rfloor}\binom{k}{4} = \begin{cases} \dfrac{k-1}{2}(k(q-1) - (k-2)(k-3)), & k \text{ even}, \\ \dfrac{k}{2}((k-1)(q-1) - (k-2)(k-3)), & k \text{ odd}. \end{cases} \tag{16}$$

For comparison, some values are

q	5	8	9	11	13	16	17	19
$L_1(q)$	4	5	5	6	6	7	7	7
$L_2(q)$	4	6	6	6	7	8	8	8
$\rho_{\mathcal{C}}(q)$	4	6	6	6	8	9	9	10

The last row combines the analytic lower bounds at $q = 5, 8, 9, 11$, the kernel-checked classification at $q = 16$, and the exhaustive classifications at $q = 13, 17, 19$ described in Section D. Thus the second-moment correction alone proves, for example, that a $\mathcal{C}$-complete arc in $\mathrm{PG}(2, 16)$ has at least eight points, while the finite classifications determine the displayed exact values.

The routine algebra behind the additive bound is isolated in the next lemma on the full interval needed to avoid any parity split.

Lemma 4.8 (Polynomial estimate)**.** *For $s \geq 2$ and $0 \leq a \leq 2$, put*

$$P_s(a) = \frac{2a-3}{4}s^3 + \frac{a^2 - 7a + 10}{4}s^2 + \frac{-3a^2 + 10a - 8}{2}s + \frac{-a^3 + 5a^2 - 8a + 6}{2}.$$

If $P_s(a) \geq 0$, then $a \geq 3/2 - 8/s$.

Proof. On $0 \leq a \leq 2$, the coefficients of s^2, s, and 1 in $P_s(a)$ are at most $5/2$, $1/6$, and 3, respectively; for the middle coefficient, $1/6 - (-3a^2 + 10a - 8)/2 = (3a - 5)^2/6$. Since $s \geq 2$, their total contribution is at most

$$\frac{5}{2}s^2 + \frac{1}{6}s + 3 \leq 4s^2.$$

Thus $P_s(a) \geq 0$ implies $0 \leq (2a - 3)s^3/4 + 4s^2$, and division by s^2 gives the claim. □

For ordinary complete arcs, the leading $\sqrt{2q}$ scale is classical; compare Ball [9, Theorem 3.1]. The theorem below obtains the displayed additive refinement for conic-relative completeness from the prescribed-hole correction.

Theorem 4.9 (Explicit additive lower bound)**.** *For every prime power q,*

$$\boxed{\rho_{\mathcal{C}}(q) \geq \sqrt{2q} + \frac{3}{2} - \frac{8}{\sqrt{2q}}.} \tag{17}$$

Consequently, as q tends to infinity through prime powers,

$$\liminf_{q \to \infty} (\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2}. \tag{18}$$

Proof. For $q = 2$ the right side of (17) is negative, so the assertion is immediate. Assume $q \geq 3$. Let k be the size of a $\mathcal{C}$-complete arc and put $s = \sqrt{2q}$. The elementary first-moment inequality $q^2 - k \leq \binom{k}{2}(q-1)$ implies $k \geq s$ for $q \geq 2$: otherwise $k < s \leq q$, while $k^2 < 2q$ gives $\binom{k}{2} < q$, so its right side is smaller than $q(q-1)$ whereas $q^2 - k \geq q(q-1)$. If $k \geq s+2$, then (17) is immediate. Otherwise write $k = s + a$, so $0 \leq a < 2$ and $s \geq 2$.

Since $m = \lfloor k/2 \rfloor \leq k/2$,

$$\frac{6}{m}\binom{k}{4} \geq \frac{12}{k}\binom{k}{4} = \frac{(k-1)(k-2)(k-3)}{2}.$$

Dropping the nonnegative conic term in (14) therefore gives the necessary inequality

$$q^2 - k \le \frac{k-1}{2}(k(q-1) - (k-2)(k-3)).$$

After substituting $q = s^2/2$ and $k = s + a$, the right side minus the left side is

$$\frac{s+a-1}{2}\left((s+a)\left(\frac{s^2}{2}-1\right) - (s+a-2)(s+a-3)\right) + (s+a) - \frac{s^4}{4}$$
$$= \frac{2a-3}{4}s^3 + \frac{a^2-7a+10}{4}s^2 + \frac{-3a^2+10a-8}{2}s + \frac{-a^3+5a^2-8a+6}{2} = P_s(a).$$

It is nonnegative, so Lemma 4.8 gives $a \ge 3/2 - 8/s$, which is (17). Letting q tend to infinity through prime powers gives (18). $\square$

The conic supplies the finite bound (17), but the limit (18) needs far fewer than $q+1$ exempt points to be disturbed: the hole count must reach the scale $q^{3/2}$ before the additive constant moves.

Theorem 4.10 (Large prescribed hole sets)**.** *For each q in an infinite set of prime powers, let $\mathcal{H}_q$ be a set of h_q points of a projective plane of order q and let A_q be a k_q-arc disjoint from $\mathcal{H}_q$ and complete outside $\mathcal{H}_q$. If $h_q/q^{3/2} \to \lambda$, then*

$$\liminf_{q\to\infty}(k_q - \sqrt{2q}) \ge \frac{3}{2} - \frac{\lambda}{\sqrt{2}}. \tag{19}$$

In particular the constant $3/2$ of (18) persists whenever $h_q = o(q^{3/2})$.

Proof. Drop the nonnegative term $I_{\mathcal{H}}(A_q)/m$ from (8):

$$q^2 + q + 1 - k_q - h_q \le \binom{k_q}{2}(q-1) - \frac{6}{m}\binom{k_q}{4}.$$

The first-moment inequality $q^2 + q + 1 - k_q - h_q \le \binom{k_q}{2}(q-1)$ with $h_q = O(q^{3/2})$ gives $k_q^2 \ge 2q - O(q^{1/2})$, so $a_q = k_q - \sqrt{2q}$ is bounded below. Along any subsequence on which a_q is bounded above, write $k = \sqrt{2q} + a$ with a in a fixed bounded set. Then $\binom{k}{2}(q-1) = q^2 + (a - \frac{1}{2})\sqrt{2}\,q^{3/2} + O(q)$ and, for either parity of k, $\frac{6}{m}\binom{k}{4} = \frac{1}{2}k^3 + O(k^2) = \sqrt{2}\,q^{3/2} + O(q)$, with the $O(q)$ terms uniform in a. The displayed inequality becomes

$$0 \le \sqrt{2}\Big(a - \frac{3}{2}\Big)q^{3/2} + h_q + O(q),$$

and dividing by $\sqrt{2}\,q^{3/2}$ gives $a \ge 3/2 - \lambda/\sqrt{2} - o(1)$. Subsequences on which $a_q \to \infty$ satisfy (19) trivially. $\square$

No property of $\mathcal{H}_q$ beyond its size is used. A line, a conic, a unital, or any $O(q^{3/2-\varepsilon})$ points may be exempted without loss in the constant, while exempting $\lambda q^{3/2}$ points costs exactly $\lambda/\sqrt{2}$ in the bound stated.

Remark 4.11 (The scale of the conic term)*.* Every secant meets $\mathcal{C}$ in at most two points, so

$$0 \le I_{\mathcal{C}}(A) \le 2\binom{k}{2}.$$

At the relevant scale $k \asymp \sqrt{q}$, the term $I_{\mathcal{C}}(A)/m$ is therefore $O(\sqrt{q})$, while the universal overlap correction $6\binom{k}{4}/m$ is $\Theta(q^{3/2})$. Consequently, improving the lower bound on $I_{\mathcal{C}}(A)$ within (14) can be useful for finely balanced finite cases, but cannot improve the additive constant $3/2$ in Theorem 4.9. Any stronger asymptotic result needs an additional mechanism, not merely a spectral estimate for $I_{\mathcal{C}}(A)$.

Remark 4.12 (Ordinary completion as independent domination). A $\mathcal{C}$-complete arc is one step from an ordinary complete arc, and the step is a graph problem on the uncovered locus. Let A be $\mathcal{C}$-complete with $U = \mathcal{U}_{\mathcal{C}}(A) \subseteq \mathcal{C}$, and let Γ_A be the graph on U in which $u \sim v$ when the line uv contains a point of A. For $S \subseteq \mathcal{C}$, the set $A \cup S$ is an arc if and only if $S \subseteq U$ and S is independent in Γ_A: a covered conic point lies on a secant, three points of $\mathcal{C}$ are never collinear, and a collinear triple a, s, s' with $a \in A$ is exactly an edge ss'. Such an $A \cup S$ is an ordinary complete arc if and only if S is a maximal independent set, because every point outside U is already covered, a line ss' meets $\mathcal{C}$ only at s, s', and $u \in U \setminus S$ lies on the secant as exactly when $u \sim s$. So the least number of conic points that complete A is the independent domination number $i(\Gamma_A)$. Each $a \in A$ contributes the matching $\{u, \iota_a(u)\}$ of its chord involution restricted to U, so Γ_A is a union of k matchings, has maximum degree at most k, and $i(\Gamma_A) \geq |U|/(k+1)$.

5 Coverage by caps and the secant-local remainder

The defect identity uses three facts about secants: a line has $q+1$ points, two secants meet in at most one point, and the number of intersecting pairs of secants with disjoint endpoints is determined by the arc. Only the third fact changes for caps in higher dimension, where two such secants meet exactly when their four endpoints are coplanar. This section records the resulting identity and then extracts more from it than the planar argument does, by counting coplanar quadruples through each secant separately.

Throughout, A is a k-cap in $\mathrm{PG}(n, q)$ with $n \geq 2$ and $k \geq 4$: a set of k points, no three collinear. Put $\theta_j = (q^{j+1} - 1)/(q-1)$, keep $N = \binom{k}{2}$, $m = \lfloor k/2 \rfloor$, and $r(x)$ for the number of secants through a point $x \notin A$, and let $\mathcal{H}$, $V_{\mathcal{H}}(A)$, $\mathcal{X}_{\mathcal{H}}(A)$, and $I_{\mathcal{H}}(A)$ be as in Section 3. Let $c_4(A)$ be the number of four-subsets of A that lie in a plane. Lemma 2.1 and its proof hold verbatim, so $r(x) \leq m$.

Theorem 5.1 (Cap defect identity)**.** *For every k-cap A in* $\mathrm{PG}(n, q)$*,*

$$\sum_{x \notin A} r(x) = N(q-1), \qquad \sum_{x \notin A} \binom{r(x)}{2} = 3c_4(A),$$

and with

$$\Delta_{\mathcal{H}}(A) = N(q-1) - \frac{6}{m} c_4(A) - \frac{I_{\mathcal{H}}(A)}{m} - |\mathcal{X}_{\mathcal{H}}(A)|$$

the identity (5) *holds unchanged: $m\Delta_{\mathcal{H}}(A) = \sum_{x \in \mathcal{X}_{\mathcal{H}}(A)} (r(x)-1)(m-r(x)) + \sum_{y \in \mathcal{H}} r(y)(m - r(y)) \geq 0$. In particular, a cap complete outside $\mathcal{H}$ with $|\mathcal{H}| = h$ satisfies*

$$\theta_n - k - h \leq N(q-1) - \frac{6}{m} c_4(A) - \frac{I_{\mathcal{H}}(A)}{m}. \tag{20}$$

Proof. The first equation counts incidences as in Proposition 2.2. For the second, two secants through a common point $x \notin A$ have four distinct endpoints by the cap analogue of Lemma 2.1, and two secants with disjoint endpoints meet if and only if they span a plane, that is, if and only if their four endpoints are coplanar; the meeting point is outside A since no three cap points are collinear. A coplanar four-subset has three pairings, each giving one intersecting pair. The proof of Theorem 3.1 used only these two equations and $r \leq m$, so it applies with $3\binom{k}{4}$ replaced by $3c_4(A)$. $\square$

For $n = 2$ every four-subset is coplanar and $c_4(A) = \binom{k}{4}$. For $n \geq 3$ the nonnegative summands have the same stability content as before, each nonzero one being at least $m -$

2, but two features of the plane are lost: $D(A)$ need not be an integer, so the gap of Proposition 3.6 does not transfer, and secants with disjoint endpoints may be skew, so the Kneser graph of Theorem 3.7 no longer decomposes into concurrence cliques. What is gained is a term, $c_4(A)$, that the geometry constrains.

5.1 Coplanar quadruples through a secant

Fix a secant ℓ with endpoints a, b and let

$$T_\ell = \#\{B \in \tbinom{A}{4} : a, b \in B,\ B \text{ coplanar}\}.$$

The $t = \theta_{n-2}$ planes through ℓ partition $A \setminus \{a, b\}$; if the plane π contains z_π of these points, then $\{a, b, c, d\}$ is coplanar exactly when c, d lie in the same plane through ℓ, so

$$T_\ell = \sum_{\pi \supset \ell} \binom{z_\pi}{2}, \qquad \sum_{\pi \supset \ell} z_\pi = k - 2, \qquad \sum_{\ell} T_\ell = 6c_4(A). \tag{21}$$

The same number counts the secants that meet ℓ outside A, one for each coplanar $\{c, d\}$, and a point $x \in \ell \setminus A$ accounts for $r(x) - 1$ of them:

$$T_\ell = \sum_{x \in \ell \setminus A} (r(x) - 1). \tag{22}$$

Lemma 5.2 (Pencil bound)**.** *Write $k-2 = ts+b$ with $0 \le b < t$ and put $B_n(k, q) = t\binom{s}{2}+sb$. Then for every secant,*

$$T_\ell = B_n(k, q) + \frac{1}{2} \sum_{\pi \supset \ell} (z_\pi - s)(z_\pi - s - 1), \tag{23}$$

where every summand is a nonnegative integer. Hence $T_\ell \ge B_n(k, q)$, with equality exactly when every plane through ℓ contains s or $s+1$ further cap points, and $c_4(A) \ge \frac{1}{6}\binom{k}{2} B_n(k, q)$.

Proof. For integers z and s, $\binom{z}{2} - sz + \binom{s+1}{2} = \frac{1}{2}(z - s)(z - s - 1)$, and the product of two consecutive integers is nonnegative. Sum over the t planes using $\sum z_\pi = k - 2 = ts + b$: $s(k - 2) - t\binom{s+1}{2} = t\binom{s}{2} + sb$. ☐

Substituting (23) into Theorem 5.1 gives a three-term identity whose summands are all nonnegative: concurrence multiplicities away from 1 and m, holes of intermediate index, and unbalanced pencils. In the plane $t = 1$, $s = k - 2$, and the pencil term vanishes, so $NB_2(k, q) = 6\binom{k}{4}$ recovers (5).

5.2 A secant-local coverage bound

Replacing $c_4(A)$ by its lower bound in (20) is not the best use of (22). The intersections on a secant cost coverage in a way that depends on how they are distributed along it, and the cost is smallest when they are concentrated.

For $m \ge 2$ and an integer $T \ge 0$, write $T = j(m - 1) + \rho$ with $0 \le \rho < m - 1$ and put

$$\varphi_m(T) = j\,\frac{m - 1}{m} + \frac{\rho}{\rho + 1}.$$

Thus $\varphi_m(T) = T/(T + 1)$ for $T \le m - 1$, $\varphi_m(T) = T/m$ when $m - 1$ divides T, and $\varphi_m(T) \ge \max\{T/m,\ T/(T + 1)\}$ always.

Theorem 5.3 (Secant-local coverage bound)**.** *For every k-cap A in* $\mathrm{PG}(n,q)$ *and every* $\mathcal{H} \subseteq \mathrm{PG}(n,q) \setminus A$*,*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N(q-1) - \sum_{\ell} \varphi_m(T_\ell) - \frac{I_{\mathcal{H}}(A)}{m}, \tag{24}$$

the sum running over the secants of A. Consequently

$$|\mathcal{X}_{\mathcal{H}}(A)| \le N\Big(q - 1 - \varphi_m(B_n(k,q))\Big) - \frac{I_{\mathcal{H}}(A)}{m}. \tag{25}$$

Proof. Let Y be the set of points outside A on at least one secant. Then $N(q-1) - |Y| = \sum_{x \in Y}(r(x) - 1) = \sum_\ell L_\ell$ with

$$L_\ell = \sum_{x \in \ell \setminus A} \Big(1 - \frac{1}{r(x)}\Big),$$

since a point of index r receives $(r-1)/r$ from each of its r secants. On a fixed secant the indices $r(x)$, $x \in \ell \setminus A$, are integers in $[1,m]$ with $\sum(r(x)-1) = T_\ell$ by (22). The function $r \mapsto 1 - 1/r$ is concave, so the minimum of L_ℓ over the polytope of real vectors with these constraints is attained at a vertex, where at most one coordinate lies strictly between 1 and m. Coordinates equal to 1 contribute nothing, j coordinates equal to m contribute $j(m-1)/m$ and use $j(m-1)$ of the total, and the remaining coordinate $\rho + 1$ contributes $\rho/(\rho+1)$; so $L_\ell \ge \varphi_m(T_\ell)$. Finally $|\mathcal{X}_{\mathcal{H}}(A)| = |Y| - \#\{y \in \mathcal{H} : r(y) > 0\}$ and each covered hole has index at most m, so at least $I_{\mathcal{H}}(A)/m$ holes are covered. The explicit form follows from Lemma 5.2 and monotonicity of φ_m. □

The term T_ℓ/m in φ_m reproduces the correction $6c_4(A)/m$ of Theorem 5.1; the term $T_\ell/(T_\ell+1)$ is new. It says that a secant meeting many others loses almost one point of coverage however concentrated those meetings are, with equality when they all occur at a single point.

5.3 Three dimensions

In $\mathrm{PG}(3,q)$ a secant lies in $t = q+1$ planes, so for $q+3 \le k \le 2q+4$ the balanced occupancies are 1 and 2 and $B_3(k,q) = k - q - 3$. A pair $\{a,b\}$ with $T_{ab} = 0$ is a *free pair* in the sense of Farr and Lisoněk [6], a pair not participating in any coplanar quadruple; a cap of $\mathrm{PG}(3,q)$ with a free pair has at most $q+3$ points [6, Theorem 2.3], and the pencil count quantifies what happens above that threshold: every secant meets at least $k-q-3$ others. The threshold needs no upper restriction: $\binom{z}{2} \ge z - 1$ for every integer $z \ge 0$, so $T_\ell \ge (k-2) - (q+1)$ whenever $k \ge q+3$.

Corollary 5.4 (Complete caps in $\mathrm{PG}(3,q)$)**.** *Let A be a k-cap in* $\mathrm{PG}(3,q)$ *with $k \ge q+3$ and let* $\mathcal{H} \subseteq \mathrm{PG}(3,q) \setminus A$ *with $|\mathcal{H}| = h$. Then*

$$|\mathcal{X}_{\mathcal{H}}(A)| \le \binom{k}{2}\Big(q - 2 + \frac{1}{k-q-2}\Big) - \frac{I_{\mathcal{H}}(A)}{m}, \tag{26}$$

and if A is complete outside $\mathcal{H}$,

$$q^3 + q^2 + q + 1 - k - h \le \binom{k}{2}\Big(q - 2 + \frac{1}{k-q-2}\Big) - \frac{I_{\mathcal{H}}(A)}{m}. \tag{27}$$

If $h_q = \lambda q^2 + o(q^2)$ along a sequence of caps complete outside $\mathcal{H}_q$, then

$$\liminf_{q \to \infty} (k_q - \sqrt{2}\,q) \ge \frac{1}{2} + \frac{3-\lambda}{\sqrt{2}}. \tag{28}$$

For ordinary complete caps ($\lambda = 0$) the constant is $\frac{1}{2} + \frac{3}{\sqrt{2}} = 2.621\ldots$, and for caps complete outside a nonsingular quadric, which has q^2+1 or $(q+1)^2$ points ($\lambda = 1$), it is $\frac{1}{2} + \sqrt{2}$.

Proof. For $k \geq q+3$, $T_\ell \geq k-q-3$ on every secant and $\varphi_m(T) \geq T/(T+1)$ is nondecreasing in T, so (24) gives $L \geq N\frac{k-q-3}{k-q-2} = N(1 - \frac{1}{k-q-2})$, which is (26); completeness turns it into (27). For the limit, drop $I_{\mathcal{H}}/m$ and note that the first-moment inequality $\theta_3 - k - h \leq N(q-1)$ bounds $a_q = k_q - \sqrt{2}\,q$ from below. Along a subsequence with a bounded, $N = q^2 + \sqrt{2}\,a\,q - \frac{\sqrt{2}}{2}q + O(1)$ and $1/(k-q-2) = O(1/q)$, so the right side of (27) is $q^3 + (\sqrt{2}\,a - \frac{\sqrt{2}}{2} - 2)q^2 + O(q)$, while the left side is $q^3 + (1-\lambda)q^2 + o(q^2)$. Comparing the q^2 coefficients gives $\sqrt{2}\,a \geq 3 - \lambda + \frac{\sqrt{2}}{2} - o(1)$. $\square$

The first moment alone gives the constant $\frac{1}{2} + \sqrt{2} = 1.914\ldots$ for ordinary complete caps, so the secant-local term contributes exactly $1/\sqrt{2}$. The finite inequality is usable directly. At $q = 3$ it reads $40 - k \leq \binom{k}{2}(1 + \frac{1}{k-5})$, which fails for $k = 7$ ($33 > 31.5$) and holds for $k = 8$, whereas the first moment allows $k = 7$. At $q = 13$ and $k = 21$ it reads $2359 \leq 2345$, so no complete 21-cap exists in $\mathrm{PG}(3,13)$, while the first moment permits one. The smallest complete caps of $\mathrm{PG}(3,q)$ have 8, 12, and 17 points for $q = 3, 5, 7$, and the smallest known have 30 and 36 points for $q = 11, 13$ [8]; the inequality gives 8, 10, 13, 19, 22, tight at $q = 3$ and loose thereafter. The same source calls $\sqrt{2}\,q^{(n-1)/2}$ the trivial lower bound for complete caps of $\mathrm{PG}(n,q)$ and cites no improvement, and we found none in the sources we could consult; the additive term in (28) is, to our knowledge, the first refinement of the counting bound for complete caps in $\mathrm{PG}(3,q)$.

5.4 Four and more dimensions

At the covering scale $k \asymp q^{(n-1)/2}$ a secant lies in $t = \theta_{n-2} \asymp q^{n-2}$ planes, and $k - 2 < t$ for large q as soon as $n \geq 4$. Then $s = 0$, $B_n(k,q) = 0$, and Lemma 5.2 says nothing. This is not an artefact of the method. A parity-check matrix of a perfect two-error-correcting code has columns forming a cap with no coplanar quadruple, since any four columns are independent, that is complete, since every syndrome is a combination of at most two columns. The ternary Golay code gives a complete 11-cap in $\mathrm{PG}(4,3)$ and the binary repetition code of length five gives the frame in $\mathrm{PG}(3,2)$; in both, every point outside the cap lies on exactly one secant. These two are also the equality cases of Pavese's packing bound for sets with no four coplanar points [7, Proposition 3.1]. Normal rational curves, complete or not, also have $c_4(A) = 0$. Any lower bound on $c_4(A)$ for complete caps in $\mathrm{PG}(n,q)$, $n \geq 4$, must exclude these.

The invariant has a coding-theoretic reading. If H has the cap points as columns and $C = \ker H$, then a coplanar quadruple spans a one-dimensional space of dependencies with all four coefficients nonzero, because no three cap points are collinear; conversely every weight-four codeword arises this way. Hence

$$A_4(C) = (q-1)\,c_4(A), \qquad (q-1)\,T_{ab} = \#\{c \in C : \mathrm{wt}(c) = 4,\ a,b \in \mathrm{supp}(c)\},$$

so Theorem 5.3 is a statement about how the weight-four supports of C are distributed over pairs of coordinates, not only about their number.

Theorem 5.3 also isolates what a higher dimensional improvement would need. If a proportion η of the secants have $T_\ell \geq d$, then for a cap complete outside $\mathcal{H}$

$$\theta_n - k - h \leq N\Big(q - 1 - \eta\frac{d}{d+1}\Big) - \frac{I_{\mathcal{H}}(A)}{m},$$

and with $h = \lambda q^{n-1} + o(q^{n-1})$, $n \geq 4$, and $d = d(q) \to \infty$ on a proportion $\eta - o(1)$ of secants, the expansion used for Corollary 5.4, in which the term $-k/2$ of N is now of lower order, gives

$$\liminf_{q\to\infty} \frac{k - \sqrt{2}\, q^{(n-1)/2}}{q^{(n-3)/2}} \geq \frac{2 - \lambda + \eta}{\sqrt{2}},$$

against $(2-\lambda)/\sqrt{2}$ from the first moment. Since the average of T_ℓ is $6c_4(A)/N$, a lower bound $\mathbb{E}\, T_\ell \geq \gamma q$ together with $\mathbb{E}\, T_\ell^2 \leq Cq^2$ would give $\Pr(T_\ell \geq \gamma q/2) \geq \gamma^2/(4C)$ by Cauchy–Schwarz, which is such a proportion. The higher-dimensional problem is therefore to force coplanar quadruples that are both numerous and spread over the secants; it is stated as a problem in the conclusion.

6 Conclusion and further questions

The prescribed-hole identity retains the local equality and stability data discarded by the usual scalar secant count. At zero defect those local conditions assemble into a matching design. For any $q+1$ prescribed holes they sharpen the scalar lower bound; for a conic, the same mechanism also restricts equality and isolates the finite obstruction at $q = 16$. The appendices record secondary consequences and the exact formal and computational trust boundaries.

Problem 6.1. Construct $\mathcal{C}$-complete arcs of size $O(\sqrt{q})$, or prove that this is impossible for an infinite family. The upper-bound transfer in Appendix B currently supplies only $O(\sqrt{q}(\log q)^c)$. On the lower-bound side, Remark 4.11 shows that improving the additive constant $3/2$ requires a mechanism coupling rank-three realization to the matching structure, rather than a sharper estimate of the conic-incidence term. The identity (5) makes the target exact. For $k = \sqrt{2q} + O(1)$ the defect of a $\mathcal{C}$-complete arc is $\sqrt{2}(k - \sqrt{2q} - \frac{3}{2})q^{3/2} + O(q)$, so any improvement of the constant needs a defect of order $q^{3/2}$; excluding zero defect, or proving a bounded positive defect, cannot help. A sufficient geometric condition: if the points x with $r(x) \leq (1-\varepsilon)m$ carry a proportion η of the pairs of concurrent secants, then $(r-1)(m-r) \geq \frac{2\varepsilon}{1-\varepsilon}\binom{r}{2}$ at each of them gives $m\Delta_{\mathcal{C}}(A) \geq \frac{6\eta\varepsilon}{1-\varepsilon}\binom{k}{4}$ and hence $\liminf(\rho_{\mathcal{C}}(q) - \sqrt{2q}) \geq \frac{3}{2} + \frac{\eta\varepsilon}{1-\varepsilon}$. Whether rank-three geometry forces such a proportion is open.

Problem 6.2. In $\mathrm{PG}(n,q)$ with $n \geq 4$, the pencil bound of Section 5 is vacuous at the covering scale, and complete caps with no coplanar quadruple exist. Find a lower bound on the number $c_4(A)$ of coplanar quadruples of a complete cap at $k \asymp q^{(n-1)/2}$ together with an upper bound on its concentration over secants; by Theorem 5.3 a proportion η of secants with unbounded T_ℓ improves the second-order term of the lower bound by $\eta/\sqrt{2}$.

Problem 6.3. Determine whether the secant-local bound of Corollary 5.4 is attained. Equality forces the secants of a complete cap in $\mathrm{PG}(3,q)$ to split into concurrent bundles of size $k-q-2$ with secants from different bundles skew, every plane meeting the cap in at most four points, and $\binom{k}{2}(q - 2 + \frac{1}{k-q-2}) = q^3 + q^2 + q + 1 - k$; for $k = q+4$ the last condition fails for every q.

A Equality classifications

The first two nontrivial orders illustrate the distinction between an abstract matching design and its rank-three projective realization. This secondary classification is not needed for the defect identity, the conic lower bound, or the $q = 16$ exclusion. Its role is to show that abstract matching-design existence is weaker than projective realizability, which can impose field and characteristic restrictions.

Definition A.1. A *rank-three projective realization* of a $\mathrm{MATCH}(k, m, 1)$ design $\mathcal{D}$ over a field K is an injective assignment $i \mapsto p_i \in \mathrm{PG}(2, K)$ whose image is a k-arc and such that, for every matching $M \in \mathcal{D}$, the m lines $p_i p_j$, $\{i, j\} \in M$, are concurrent. Two matching designs are *isomorphic* when a permutation of their k vertices carries the matching collection of one onto that of the other.

Corollary A.2 (Six and seven points)**.** *(a) In a Desarguesian plane, if a six-arc has every intersection of two disjoint secants on a third secant, then the field has characteristic two, contains* $\mathbb{F}_4$*, and the arc is projectively equivalent to*

$$(1,0,0),\ (0,1,0),\ (0,0,1),\ (1,1,1),\ (1,\omega,\omega^2),\ (1,\omega^2,\omega), \qquad \omega^2+\omega+1=0.$$

Conversely, this configuration has the stated concurrence property. In particular, every zero-defect six-arc in a Desarguesian plane has this form.

(b) In any projective plane, every seven-arc has $\Delta_{\mathcal{H}}(A) \geq 2$ *with respect to every prescribed hole set.*

Proof. For six points, the three intersections of opposite sides of any complete quadrangle must all lie on the secant through the remaining pair. Normalize four vertices to

$$(1,0,0), \quad (0,1,0), \quad (0,0,1), \quad (1,1,1).$$

Their diagonal points are

$$(1,1,0), \qquad (1,0,1), \qquad (0,1,1).$$

Their determinant is -2, so they are collinear exactly in characteristic two, on $x+y+z=0$; that line contains the remaining pair. Write a fifth point as $(1, t, 1+t)$. Applying the diagonal-line condition to the quadrangle formed by the first three frame points and this fifth point gives the diagonal line

$$t(1+t)x + (1+t)y + tz = 0.$$

It must contain the remaining frame point $(1,1,1)$, which gives $t^2+t+1=0$; its intersection with $x+y+z=0$ is the sixth point $(1, t^2, t)$. Direct substitution verifies all required concurrences.

For seven points, Theorem 3.7 would produce a $\mathrm{MATCH}(7,3,1)$ design. Alspach and Heinrich [3, Theorem 3.1] proved that no such design exists. Here $v = 35$, so Corollary 3.10 gives the stated quantitative bound. □

Following Alspach and Heinrich's hyperoval construction [3, Theorem 1.2], call the $\mathrm{MATCH}(10,5,1)$ class formed by the 63 concurrence matchings of the regular hyperoval in $\mathrm{PG}(2,8)$ the *regular-hyperoval design.* It is one of Mathon's two abstract classes.

Theorem A.3 (Rank-three ten-point classification)**.** *Let K be a field. Among the simple* $\mathrm{MATCH}(10,5,1)$ *designs, only the regular-hyperoval design can have a rank-three projective realization. It has one over K if and only if*

$$\operatorname{char} K = 2, \qquad \textit{and} \qquad \mathbb{F}_8 \subseteq K.$$

The other abstract design has no rank-three realization over any field.

The proof keeps three layers separate. Mathon's theorem supplies the two abstract matching-design classes; the partial-geometry calculation translates their incidence parameters; and a normalized projective realization is then tested by exact elimination. Only the last layer is computer-assisted.

Proof. Alspach and Heinrich allow repeated matchings in their definition, but when $\lambda = 1$ a repeated five-matching would repeat each of its pairs of edges and is therefore impossible. Thus their statement that Mathon found precisely two nonisomorphic $\mathrm{MATCH}(10, 5, 1)$ designs applies to Definition A.1 [3, p. 40, opening classification paragraph]. Each matching design has an associated partial geometry: take $E(K_{10})$ as points and the 63 matching blocks as lines. A line has five points and each point lies on seven lines. If an edge e is not in a perfect matching M, exactly three edges of M are disjoint from e. For each such edge, the $\lambda = 1$ property gives a unique block through it and e, and these three blocks are distinct. Thus the geometry is denoted $\mathrm{pg}(5, 7, 3)$ in Reichard and Woldar's (line size, point degree, α) convention, or $\mathrm{pg}(4, 6, 3)$ in the common $\mathrm{pg}(s, t, \alpha)$ convention.

Reichard and Woldar report Mathon's full two-class result and construct the corresponding partial geometries in Proposition 5.1.4; their Proposition 5.1.5 computes automorphism groups, and Corollary 5.1.6 distinguishes the two models [22, Proposition 5.1.4, Proposition 5.1.5, and Corollary 5.1.6]. Mathon's published two-class completeness [19], as reported both there and by Alspach–Heinrich, is an external input. The paper-local enumeration in Appendix F reconstructs and checks representatives of the two classes, but is not used to prove that the list is complete.

Here is the realization test. Relabel and normalize four vertices to

$$p_0 = (1, 0, 0), \quad p_1 = (0, 1, 0), \quad p_2 = (0, 0, 1), \quad p_3 = (1, 1, 1).$$

No remaining point lies on p_0p_1, so write $p_i = (x_i, y_i, 1)$ for $4 \le i \le 9$. For each matching block, the first two secants determine its proposed concurrence point; requiring each of the other three secants to pass through that point gives three determinant equations. The 63 blocks therefore give 189 integral equations, and together with the arc inequations they are necessary and sufficient for a realization.

Only a necessary part of the arc open set is needed for the exclusions. Indeed,

$$\det(p_1, p_2, p_9) = x_9, \qquad \det(p_1, p_3, p_9) = x_9 - 1,$$

so every arc realization has $x_9(x_9 - 1) \neq 0$. The certificate saturates the concurrency ideal by this product. This enlarges, rather than shrinks, the locus that must be excluded: a unit saturated ideal therefore rules out every arc realization without claiming that this one product encodes all arc inequations.

The remaining elimination is computer-assisted. The exact certificate summarized in Appendix F.2 gives the following mathematical output. After inverting 2, rational module identities make both saturated ideals the unit ideal, and their only denominator prime is 2; hence both designs are excluded in every odd characteristic. In characteristic two the second design again has unit saturated ideal. For the regular-hyperoval design, the surviving triangular basis forces a parameter t with $t^3 + t + 1 = 0$, while the arc inequations force $t \notin \mathbb{F}_2$. Thus $\mathbb{F}_8 \subseteq K$. The regular hyperoval over $\mathbb{F}_8$ and scalar extension prove the converse. □

The certificate says more than existence. In characteristic two the lexicographic basis for the regular-hyperoval design is triangular: its univariate generator is $t(t+1)(t^3+t+1)$ in the parameter t, the next generator fixes one coordinate as t^3, and every remaining generator is linear in a fresh unknown. The arc inequations remove $t \in \mathbb{F}_2$, so the normalized realizations are exactly three points over $\overline{\mathbb{F}}_2$, with coordinates in $\mathbb{F}_8$, forming one orbit of the Frobenius map $t \mapsto t^2$.

Corollary A.4 (Projective uniqueness of the ten-point realization). *Every rank-three projective realization of the regular-hyperoval design over a field K is carried by an element of* $\mathrm{PGL}(3, K)$ *onto the regular hyperoval of* $\mathrm{PG}(2, 8)$*, embedded in* $\mathrm{PG}(2, K)$ *through* $\mathbb{F}_8 \subseteq K$.

Proof. By Theorem A.3, K has characteristic two and contains $\mathbb{F}_8$. A projectivity moves the four frame points of any realization to their normalized positions, so every realization is $\mathrm{PGL}(3, K)$-equivalent to one of the three normalized solutions. The regular hyperoval of $\mathrm{PG}(2, 8)$, labelled as in the converse above, normalizes to one of them, and the other two are its images under the Frobenius collineations of $\mathrm{PG}(2, 8)$. A Frobenius collineation carries a conic and its nucleus to a conic and its nucleus, and all conics of $\mathrm{PG}(2, 8)$ are projectively equivalent, so each of the three solutions is a regular hyperoval as a point set. □

The corollary classifies the arc, not the labelling and not the prescribed conic: the 63 matchings may be attached to the ten points in several ways, and the disjoint conic of the relative-completeness problem is additional data.

Away from characteristic two, a single classical inequality settles every size at once. Hirzebruch [4] proved, by way of the Kummer covers of the plane branched along a line arrangement, that an arrangement of $d \geq 6$ lines in $\mathrm{PG}(2, \mathbb{C})$ with $t_d = t_{d-1} = t_{d-2} = 0$ satisfies

$$t_2 + \frac{3}{4} t_3 \geq d + \sum_{r \geq 5} (2r - 9) t_r, \tag{29}$$

where t_r is the number of points lying on exactly r of the lines; this is the form recorded by Pokora [5, Remark 2.9], and the weaker form $t_2 + t_3 \geq d + \sum_{r \geq 5} (r - 4) t_r$ holds already for $d \geq 4$ and $t_d = t_{d-1} = 0$ [5, Theorem 1.5].

Theorem A.5 (No realizations in characteristic zero). *Let $k \geq 6$ and $m = \lfloor k/2 \rfloor$. No simple* $\mathrm{MATCH}(k, m, 1)$ *design has a rank-three projective realization over a field of characteristic zero.*

Proof. Let K have characteristic zero and suppose a realization exists. Its finitely many coordinates generate a finitely generated subfield of K, which embeds in $\mathbb{C}$; incidences and non-incidences are polynomial equations and inequations preserved by the embedding, so we may take $K = \mathbb{C}$.

Consider the arrangement of the $d = \binom{k}{2}$ secant lines. The arc points lie on $k - 1$ secants each, and no other secant passes through an arc point because the points form an arc. Two secants with disjoint endpoints meet at a point outside A; the unique block containing the pair is concurrent there by Definition A.1, so the point lies on those $m = \lfloor k/2 \rfloor$ secants, and on no further secant, since Lemma 2.1 holds in every projective plane. Hence the multiple points are the k arc points, each of multiplicity $k - 1$, and the block points, each of multiplicity m; their number is

$$t_m = \frac{\binom{d}{2} - k\binom{k-1}{2}}{\binom{m}{2}},$$

since every pair of disjoint secants lies in exactly one block. Every multiplicity is at most $k - 1 < d - 2$, so (29) applies.

For $k \geq 8$ one has $m \geq 4$ and $k - 1 \geq 7$, so $t_2 = t_3 = 0$ and the left side of (29) vanishes while the right side is positive. For $k = 6$: $d = 15$, $t_5 = 6$, $t_3 = 15$, and (29) would read $45/4 \geq 21$. For $k = 7$: $d = 21$, $t_6 = 7$, $t_3 = 35$, and it would read $105/4 \geq 42$. Each is false. □

For $k = 7$ the weaker form gives $35 \geq 35$ and decides nothing, so the three-quarters coefficient is used. The case $k = 6$ is instructive: the fifteen block points and fifteen secants form the duad–syntheme configuration of Cremona and Richmond, which is realizable over the reals, and the obstruction is the requirement that the six arc points carry five secants each. The design for $k = 4$, the complete quadrilateral with $d = 6$, $t_3 = 4$, $t_2 = 3$, attains equality in (29). Together with Corollary A.2 and Theorem A.3, the theorem shows that the realizable equality designs found so far live in characteristic two, and that the ten-point design has a single realizable characteristic.

B An upper-bound transfer from complete arcs

Let $t_2(2, q)$ denote the smallest size of an ordinary complete arc in $\mathrm{PG}(2, q)$.

Theorem B.1 (Projective averaging). *Let $H \subseteq \mathrm{PG}(2, q)$ and let B be an ordinary complete b-arc. If*

$$b|H| < q^2 + q + 1, \tag{30}$$

then some projective image of B is disjoint from H and complete outside H.

Proof. Choose a uniformly random projectivity $g \in \mathrm{PGL}(3, q)$. Point transitivity gives

$$\mathbb{E}|gB \cap H| = \frac{b|H|}{q^2 + q + 1} < 1.$$

Since the intersection size is a nonnegative integer, it is zero for some g. Projectivities preserve ordinary completeness, so gB is complete outside H. $\square$

Corollary B.2 (Conic transfer). *If $t_2(2, q) \leq q$, then*

$$\rho_{\mathcal{C}}(q) \leq t_2(2, q). \tag{31}$$

More generally, every complete b-arc with $b \leq q$ has a projective image disjoint from the prescribed conic.

Proof. Take $H = \mathcal{C}$, so $|H| = q + 1$. For integral $b \leq q$ one has $b(q + 1) < q^2 + q + 1$, and Theorem B.1 applies. $\square$

The hypothesis $t_2(2, q) \leq q$ is part of the corollary, not a uniform assertion about every order. The application below uses only the range of sufficiently large q, where the Kim–Vu bound supplies it.

Kim and Vu proved that, for all sufficiently large q, every projective plane of order q has a complete arc of size at most $\sqrt{q}(\log q)^c$ for an absolute constant c [16, Theorem 1.1]. This is less than q for large q, so Corollary B.2 gives the following immediate consequence.

Corollary B.3. *For some absolute constant c and all sufficiently large prime powers q,*

$$\rho_{\mathcal{C}}(q) \leq \sqrt{q}(\log q)^c.$$

The transfer does not exploit the conic and is unlikely to be sharp. It nevertheless supplies a uniform upper bound without having to preserve the arc condition during a separate random covering construction.

C Even characteristic and the nucleus

Assume q is even and let ν be the nucleus of $\mathcal{C}$. The nucleus is outside $\mathcal{C}$, all tangents to $\mathcal{C}$ pass through ν, and every line through ν is one of these $q+1$ tangents.

Proposition C.1 (Case $\nu \in A$)**.** *Let A be a k-arc disjoint from $\mathcal{C}$ with $\nu \in A$. Exactly $k-1$ secants of A are tangent to $\mathcal{C}$. In particular,*

$$I_{\mathcal{C}}(A) \geq k-1, \qquad I_{\mathcal{C}}(A) \equiv k-1 \pmod 2.$$

If A is $\mathcal{C}$-complete, then

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{k-1}{m}. \tag{32}$$

Proof. For every $a \in A \setminus \{\nu\}$, the line νa is tangent to $\mathcal{C}$, giving $k-1$ conic-tangent secants. No line joining two points of $A \setminus \{\nu\}$ can be tangent: every tangent contains ν, which would place three points of A on one line. Hence these are exactly the conic-tangent secants of A.

Every tangent contributes one to $I_{\mathcal{C}}(A)$, while every other line contributes zero or two. This proves the lower bound and parity statement. Inequality (32) follows from Corollary 4.1. □

Proposition C.2 (Case $\nu \notin A$)**.** *Let A be a $\mathcal{C}$-complete arc with $\nu \notin A$. Then*

$$1 \leq r(\nu) \leq m,$$

the conic-tangent secants of A are exactly the secants through ν, and

$$I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2, \qquad I_{\mathcal{C}}(A) \geq r(\nu) \geq 1.$$

Consequently,

$$q^2 - k \leq \binom{k}{2}(q-1) - \frac{6}{m}\binom{k}{4} - \frac{1}{m}. \tag{33}$$

Proof. The nucleus is a required point outside $A \cup \mathcal{C}$, so relative completeness gives $r(\nu) \geq 1$; Lemma 2.1 gives the upper bound. A line is tangent to $\mathcal{C}$ exactly when it passes through ν. Thus the number of conic-tangent secants of A is $r(\nu)$. Reducing the sum of the intersection sizes $|\ell \cap \mathcal{C}|$ modulo two gives $I_{\mathcal{C}}(A) \equiv r(\nu) \pmod 2$. The remaining claims follow from Corollary 4.1. □

Corollary C.3 (Universal even-characteristic loss)**.** *Every $\mathcal{C}$-complete arc in even characteristic satisfies*

$$I_{\mathcal{C}}(A) \geq 1.$$

Consequently, inequality (14) *can always be strengthened by replacing its final term by at least $-1/m$.*

Proof. If the nucleus belongs to A, Proposition C.1 gives $I_{\mathcal{C}}(A) \geq k-1 \geq 1$. Otherwise Proposition C.2 gives $I_{\mathcal{C}}(A) \geq r(\nu) \geq 1$. □

These two cases are useful structural reductions, but Remark 4.11 limits what the incidence term alone can accomplish. For example, at $(q,k) = (16,8)$ inequality (14) would be contradicted only by $I_{\mathcal{C}}(A) > 268$, whereas the universal upper bound is $I_{\mathcal{C}}(A) \leq 56$. Thus the exact determination of $\rho_{\mathcal{C}}(16)$ below requires information beyond a lower bound on $I_{\mathcal{C}}(A)$ inserted into the present inequality.

D Verified finite-field examples

This appendix has two independent tasks. The first combines the universal bound with explicit coordinates and trusted projective classifications to settle the small odd orders. The second proves $\rho_{\mathcal{C}}(16) = 9$ through a kernel-checked augmentation certificate. A reader following only the latter should retain three steps: the defect bound gives eight; a hypothetical $\mathcal{C}$-complete eight-arc forces its ordinary-uncovered locus onto a quadratic avoiding the arc; and the certificate excludes every such quadratic, while a displayed nine-point witness gives the upper bound.

The values at $q = 5, 8, 9, 11$ use the lower bound $L_2(q)$ and explicit coordinates; at $q = 5$ the witness is the projective four-frame. Exhaustive projective classifications reject size 7 at $q = 13$, size 8 at $q = 17$, and sizes $8, 9$ at $q = 19$; checked witnesses attain sizes $8, 9, 10$. At $q = 16$, the defect bound and a checked witness give $8 \leq \rho_{\mathcal{C}}(16) \leq 9$, and the checked covering list excludes eight.

D.1 Evaluation obstruction

The algebraic rejection used in that classification is an instance of the following elementary linear-algebra observation. Linear systems of quadrics containing arcs are an established tool; see Glynn [14] and Ball [10]. Fix nonzero vector representatives of the projective points under consideration. If V is a finite-dimensional linear system of homogeneous forms, write $\mathrm{ev}_x \in V^*$ for evaluation at the representative of x. The zero or nonzero status of this evaluation is independent of the chosen representative.

Indeed, if an arc is complete outside a conic, then its ordinary-uncovered locus lies in the zero set of the conic equation while every selected point avoids that zero set. The next lemma isolates exactly the linear-algebraic obstruction used to rule this out at $q = 16$.

Lemma D.1 (Uncovered evaluation obstruction)**.** *Let A and U be finite sets of projective points over a field K. There is no nonzero $f \in V$ such that $U \subseteq Z(f)$ and $A \cap Z(f) = \varnothing$ if either of the following holds:*

1. *the evaluation map $f \mapsto (\mathrm{ev}_u(f))_{u \in U}$ is injective;*

2. *for some $a \in A$, the functional ev_a belongs to the span of $\{\mathrm{ev}_u : u \in U\}$.*

Conversely, if K is infinite, or if $K = \mathbb{F}_q$ and $|A| \leq q$, then the absence of such an f implies one of the two conditions. The bound $|A| \leq q$ cannot be raised: the $q + 1$ lines through the origin cover $\mathbb{F}_q^2$.

Proof. If f vanishes on U, injectivity in the first case gives $f = 0$. In the second case, applying a spanning relation for ev_a to f gives $f(a) = 0$, contrary to $A \cap Z(f) = \varnothing$.

For the converse, let $W = \{f \in V : U \subseteq Z(f)\}$ and $d = \dim W$. The first condition says $d = 0$, and the second says that some ev_a vanishes on W, because W is the annihilator of $\{\mathrm{ev}_u\}$ and V is finite-dimensional. Suppose neither holds. Then $d \geq 1$ and each $\ker(\mathrm{ev}_a|_W)$, $a \in A$, is a hyperplane of W. Over an infinite field a vector space is not a finite union of proper subspaces. Over $\mathbb{F}_q$ the union of $|A| \leq q$ hyperplanes has at most $1 + q(q^{d-1} - 1) < q^d$ vectors. In either case some $f \in W$ lies outside every kernel: it vanishes on U and at no point of A. □

For $|A| \leq q$ the lemma is therefore a complete linear-algebraic test for the existence of a form in V vanishing on U and avoiding A. It says nothing about nonsingularity, which remains a separate condition when V is the space of quadratic forms.

Lemma D.2 (Line–triangle obstruction). *Suppose that $U \subseteq \mathrm{PG}(2, K)$ contains three distinct points on a line L, together with three noncollinear points outside L. Then no nonzero homogeneous quadratic vanishes on U.*

Proof. The restriction of such a quadratic to $L \cong \mathrm{PG}(1, K)$ has degree two and three distinct zeros, so it vanishes identically. Hence the equation of L divides the quadratic. The residual linear factor must vanish at the three points outside L, contradicting their noncollinearity unless that factor, and hence the quadratic, is zero. □

For the full system of degree-d forms, the evaluation functionals are the degree-d Veronese images of the points, up to nonzero scalar. Thus the first alternative says that the uncovered Veronese images span the dual coefficient space; the second supplies a selected Veronese image already in their span. In particular, for the full degree-d system it is enough to exhibit $\binom{d+2}{2}$ uncovered points whose Veronese evaluation matrix is invertible. This formulation is not specific to conics.

D.2 Small values and the classification over $\mathbb{F}_{16}$

Proposition D.3. *For the standard conic*

$$\mathcal{C} : \ XZ = Y^2$$

in the indicated projective planes,

$$\rho_{\mathcal{C}}(3) = 3, \quad \rho_{\mathcal{C}}(5) = 4, \quad \rho_{\mathcal{C}}(8) = 6, \quad \rho_{\mathcal{C}}(9) = 6, \quad \rho_{\mathcal{C}}(11) = 6,$$
$$\rho_{\mathcal{C}}(13) = 8, \quad \rho_{\mathcal{C}}(17) = 9, \quad \rho_{\mathcal{C}}(19) = 10.$$

Proof. For $q = 3$, an arc of size at most two has at most one secant and leaves at least nine points uncovered, more than a conic holds, so $\rho_{\mathcal{C}}(3) \geq 3$. The coordinate triangle has the three coordinate lines as secants, and its ordinary-uncovered locus is the four points with all coordinates nonzero, which is the nonsingular conic $V(X^2 + Y^2 + Z^2)$. An invertible coordinate change carries this conic to $XZ = Y^2$, so $\rho_{\mathcal{C}}(3) = 3$. Equation (15) gives

$$L_2(5) = 4, \qquad L_2(8) = L_2(9) = L_2(11) = 6.$$

For $q = 5$, the standard four-frame has ordinary-uncovered locus

$$V(X^2 + Y^2 + Z^2 + XY + XZ + YZ),$$

a nonsingular six-point conic. An invertible coordinate change carries this conic to $XZ = Y^2$ and the frame to the four coordinates in Appendix G; the relative certificate is checked in Lean. The appendix also lists $\mathcal{C}$-complete arcs of size six for $q = 8, 9, 11$. It also lists the attaining arcs at $q = 13, 17, 19$; their relative certificates are kernel-checked. For the reverse inequalities, the classifier enumerates all 80 projective seven-arc classes at $q = 13$, all 5441 projective eight-arc classes at $q = 17$, all 20,361 projective eight-arc classes at $q = 19$, and all 1,053,996 legal nine-arc extensions of those classes. In each case the ordinary-uncovered quadratic evaluation matrix has full rank six, excluding containment in any conic avoiding the arc. Appendix F records the exhaustive-search contract and the external-execution boundary. □

The ordinary classification of eight-arcs over $\mathbb{F}_{16}$ records arc counts, stabilizers, secant-index distributions, and conic-contained classes. The theorem below adds the datum needed here: the ordinary uncovered locus either imposes six independent quadratic conditions or forces every containing quadratic to meet the arc.

The proof uses an exhaustive covering list, not pairwise inequivalent representatives. The following contract isolates the mathematical completeness step; Appendix F records the normalized point encoding, generator, certificate checker, and trust boundary. Let F denote the standard projective four-frame.

Lemma D.4 (Augmentation certificate contract)**.** *For $4 \le i \le 8$, let $\mathcal{L}_i$ be a finite list of i-arcs with $\mathcal{L}_4 = \{F\}$. Suppose that for every $4 \le i < 8$, every $P \in \mathcal{L}_i$, and every $x \notin P$ such that $P \cup \{x\}$ is an arc, the certificate supplies a child $P' \in \mathcal{L}_{i+1}$ and an invertible linear map whose induced projectivity carries $P \cup \{x\}$ onto P'. Then every eight-arc in* $\mathrm{PG}(2, 16)$ *is projectively equivalent to a member of $\mathcal{L}_8$.*

Proof. Choose four points of the eight-arc. Every four-arc is a projective frame, so a projectivity carries those points to F. Insert the remaining four points in any order. At each step the stated certificate transports the legal insertion to the next list. Induction through the four layers gives a member of $\mathcal{L}_8$ projectively equivalent to the original arc. $\square$

Theorem D.5 (Quadratic avoidance over $\mathbb{F}_{16}$)**.** *Let A be any eight-arc in* $\mathrm{PG}(2, 16)$*, and let $U(A)$ be the points outside A that lie on no secant of A. There is no nonzero homogeneous quadratic form Q such that*

$$U(A) \subseteq Z(Q) \qquad \textit{and} \qquad A \cap Z(Q) = \varnothing.$$

The assertion includes singular quadratic forms.

Proof. Covering list. The checked list lengths are $|\mathcal{L}_5| = 4$, $|\mathcal{L}_6| = 61$, $|\mathcal{L}_7| = 454$, and $|\mathcal{L}_8| = 2633$. Every legal insertion among all 273 projective points has a checked child and projective transporter, so Lemma D.4 proves exhaustiveness. The final length agrees with the independent class count of Al-Seraji–Al-Ogali [2, Theorem 3.8.1], but that agreement is not used.

Leaf obstruction. For 2630 leaves, the uncovered locus contains three collinear points and three noncollinear points outside their line. Lemma D.2 replaces the separate quadratic matrix inversions on all these leaves.

Only three leaves lack this pattern. Using the polynomial-basis integer encoding of Appendix G, their one-dimensional quadratic kernels are generated, up to scalar, by

leaf	Q	$A \cap Z(Q)$
89	$X^2 + Y^2 + Z^2 + XY + XZ$	$\{(1, 8, 14), (1, 11, 12)\}$
90	$5XY + 4XZ + YZ$	7 of the 8 selected points
2631	$2X^2 + Y^2 + Z^2 + 5XY + 5XZ + YZ$	$\{(1, 6, 12), (1, 11, 7)\}$.

The first and third forms split respectively as

$$(X + 6Y + 6Z)(X + 7Y + 7Z) \quad \text{and} \quad 2(X + 4Y + 15Z)(X + 15Y + 4Z);$$

the middle form is nonsingular. Direct substitution gives the displayed arc intersections. Thus every quadratic containing the uncovered locus meets the arc on each exceptional leaf, which is the second alternative of Lemma D.1. Projective transport preserves the assertion, so the checked leaf result holds for every eight-arc. $\square$

Corollary D.6 (Exact relative value over $\mathbb{F}_{16}$)**.** *For every nonsingular conic $\mathcal{C} \subseteq$* $\mathrm{PG}(2, 16)$*,*

$$\rho_{\mathcal{C}}(16) = 9.$$

Proof. Equation (15) gives $L_2(16) = 8$. If a $\mathcal{C}$-complete eight-arc existed, its ordinary-uncovered locus would lie in the zero set of the nonzero quadratic defining $\mathcal{C}$, while the arc is disjoint from that zero set. This contradicts Theorem D.5. The nine-point witness in Appendix G gives the matching upper bound. $\square$

E An inverse coda: uncovered-locus reconstruction

For $q \geq 4$ this coda does not apply to the $\mathcal{C}$-complete arcs studied above. They satisfy $U(A) \subseteq \mathcal{C}$, so at most $q+1$ points are uncovered and (1) gives $q^2 - k \leq \binom{k}{2}(q-1)$. If $\binom{k}{2} \leq q$, this forces $k \geq q$, hence $\binom{k}{2} \geq \binom{q}{2} > q$ when $q \geq 4$; so $\binom{k}{2} \geq q+1$, opposite to the strict hypothesis below. The restriction is sharp. In $\mathrm{PG}(2,3)$ the coordinate triangle has the three coordinate lines as secants, so its uncovered locus is the four points with all coordinates nonzero, which is the conic $X^2+Y^2+Z^2=0$. This 3-arc is $\mathcal{C}$-complete with $\binom{3}{2} = 3 < 4 = q+1$; it is the attaining arc for $\rho_{\mathcal{C}}(3) = 3$ in Proposition D.3.

For fixed arc length and sufficiently large q, the ordinary-uncovered locus remembers its parent arc. The inverse map has two elementary stages. Put $N = \binom{k}{2}$ and $S = \mathrm{PG}(2,q) \setminus U(A)$. First recover the secant arrangement from the gap between $q+1$, the number of points on a genuine secant, and N, the largest number of points in which a nonsecant line can meet the union. Then recover the vertices from the gap between their secant multiplicity $k-1$ and the largest possible nonvertex multiplicity $\lfloor k/2 \rfloor$:

$$U(A) \longmapsto S \longmapsto \{\ell : |\ell \cap S| > N\} \longmapsto \{x : \deg(x) = k-1\} = A.$$

The strict first gap is the only large-field hypothesis.

Proposition E.1 (Uncovered-locus reconstruction)**.** *Let A and B be k-arcs in* $\mathrm{PG}(2,q)$*, where $k \geq 3$ and*

$$q+1 > \binom{k}{2}.$$

If $U(A) = U(B)$ as literal subsets of the plane, then $A = B$ as unlabelled point sets.

Proof. The complement of $U(A)$ is the union of the $\binom{k}{2}$ distinct secants of A, and likewise for B. Let ℓ be a secant of A. If no secant of B were equal to ℓ, then the secants of B would cover at most $\binom{k}{2}$ points of ℓ, one per line. This is impossible because $|\ell| = q+1 > \binom{k}{2}$ and the two secant unions are equal. Thus every secant of A is a secant of B; since the two sets have the same finite size, their secant-line sets coincide.

It remains to recover the vertices from this line arrangement. Each point of A lies on its $k-1$ incident secants. A point outside A lies on at most $\lfloor k/2 \rfloor$ secants: distinct secants through it use disjoint pairs of the k vertices and hence form a matching. Since $k-1 > \lfloor k/2 \rfloor$ for $k \geq 3$, the set A is exactly the set of points incident with $k-1$ recovered secants, and the same description recovers B. $\square$

Corollary E.2 (Canonical inverse and automorphisms)**.** *Under the hypotheses of Proposition E.1, put $S = \mathrm{PG}(2,q) \setminus U(A)$ and $N = \binom{k}{2}$. The secants of A are exactly the lines ℓ with $|\ell \cap S| > N$, and A is exactly the set of points incident with $k-1$ such lines. Moreover,*

$$\mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(U(A)) = \mathrm{Stab}_{\mathrm{P\Gamma L}_3(q)}(A).$$

Proof. Every secant lies in S and has $q+1 > N$ points. A nonsecant line meets each of the N secants in at most one point, so it contains at most N points of their union S. The vertex-recovery rule is the one proved above. Finally, every semilinear projectivity stabilizing A stabilizes $U(A)$. Conversely, if $gU(A) = U(A)$, then $U(gA) = gU(A) = U(A)$, so Proposition E.1 gives $gA = A$. $\square$

Remark E.3 (Reconstruction from a corrupted locus). The threshold in Corollary E.2 has room for errors. Let T be any point set with $|T \mathbin{\triangle} S| \leq e$. A secant meets S in all $q+1$ of its points and a non-secant line in at most N, since it meets each secant at most once; so secants meet T in at least $q+1-e$ points and non-secants in at most $N+e$. If $2e < q+1-N$, the secants are again the lines meeting T in more than $N+e$ points, and A is recovered as before. Only the projective-plane axioms enter, so this holds in every projective plane of order q.

Remark E.4 (Sharpness of the strict threshold). The strict inequality cannot in general be weakened to $q+1 \geq \binom{k}{2}$. For the projective four-frame A over $\mathbb{F}_5$, Proposition D.3 gives $U(A) = \mathcal{C}(\mathbb{F}_5)$, while $q+1 = \binom{4}{2} = 6$. The stabilizer of $\mathcal{C}$ in $\mathrm{PGL}_3(5)$ has order $|\mathrm{PGL}_2(5)| = 120$, whereas the setwise stabilizer of a projective frame has order $4! = 24$. The conic stabilizer therefore moves A through at least five distinct four-frames, all with the same literal uncovered locus $\mathcal{C}(\mathbb{F}_5)$.

F Computational and formal verification

F.1 Formal gate and evidence roles

This appendix is not part of the first-pass mathematical route. Its job is to separate three evidence roles: ordinary proofs with independent formalization, kernel-checked finite certificates, and trusted exact computation. The table below maps each claim to its role and remaining trust; the surrounding prose records the reproducible artifact identities and certificate contracts.

In the public `finitegeom` repository, the trust boundary is recorded in `trust/ARCS_COMPLETE_OUTSIDE_CONIC.md`. The matching-packing and small odd-order supplement is recorded separately in `trust/ARCS_COMPLETE_OUTSIDE_CONIC_ADDITIONS.md`. Under `RelativeConicArcs/Gates/`, the corresponding import-only modules are

`ArcsCompleteOutsideConic.lean`,
`ArcsCompleteOutsideConicAdditions.lean`.

The archived `finitegeom` release 0.2.0 has DOI `10.5281/zenodo.21664257`; the later supplement is fixed at public commit `575cf3e9`. The generated order-16 proof is distributed separately in `finitegeom-q16-certificates`, fixed at public commit `0b04429b` and pinned there to `finitegeom` commit `a7665be6`, an ancestor of the commit above. This appendix states the mathematical certificate contracts and the trust boundary; raw generated certificates, generators, and replay scripts remain in that package.

Those two gates also check the discrete gap, complete-affine equality orders, three-value equality spectra, and the standard-conic characteristic-two conclusions. The identification of an arbitrary equality oval or hyperoval with its prescribed-conic configuration remains a paper proof.

The complete `#print axioms` output for the cited gate theorems is

`propext`, `Classical.choice`, `Quot.sound`.

The checked sources contain no `sorry`, custom axiom declaration, or `native_decide`.

The table distinguishes the three evidence roles: ordinary proofs with independent formalization, kernel-checked finite certificates, and exact computer algebra outside Lean.

Claim	Certificate and checker	Remaining trust
Defect, bounds, transfer, exact reconstruction, matching rigidity, bad-concurrence stability, and the two-unit design-nonexistence gap	`Gates.ArcsCompleteOutsideConic` and `Gates.ArcsCompleteOutsideConicAdditions` in `finitegeom`; the named reconstruction and matching terminals in the two public trust documents	Lean kernel, Mathlib, and `propext`, `Classical.choice`, and `Quot.sound`
$q = 16$ exclusion	augmentation transitions, the line–triangle pattern certificate, $2630 + 3$ Lean leaf records, and `Q16QuadraticTransport`, imported by the certificate package's own gate `Gates.Q16CertificateTrust`	The theorem trusts the Lean kernel and Mathlib. Exact Python arithmetic certifies the geometric compression independently but is not a premise of the Lean theorem; generator labels and deduplication are not trusted
Exact values at $q = 13, 17, 19$	the exhaustive projective classifier and canonical JSON summaries; `Gates.ArcsCompleteOutsideConicAdditions` checks the three attaining witnesses and upper bounds	The upper bounds trust only the Lean kernel, Mathlib, and explicit finite-field reduction. The exhaustive lower classifications trust the C++ compiler and execution; the $q = 13$ lower bound has an independent frame enumeration, while the $q = 17, 19$ class reductions have no second implementation
Ten-point realization	`check_match10_rank_three.py` and its JSON certificate	Mathon's external two-class classification; Python integer arithmetic and Singular exact arithmetic; this claim is outside the Lean gate
Small-field witnesses	`ExampleChecks/Q5.lean` through `Q16.lean`	Lean kernel and the explicit finite-field implementations; no witness-search execution is trusted

For $q = 13$ and $q = 17$, the small-order classifier fixes a projective four-frame, augments by every legal point, and retains the lexicographically least image over all ordered frames. Induction on the arc size therefore retains one representative of every projective class. At $q = 19$, it tests every canonical eight-arc and every legal extension of each one; deleting one point from an arbitrary nine-arc proves that this covers every nine-arc. The companion source is `small_odd_relative_conic_classifier.cpp`; the four canonical JSON summaries and `small_odd_relative_conic.sha256` record the parameters, counts, outputs, and replay hashes. A separate frame-normalized Python enumeration rechecks the $q = 13$ lower bound. No second implementation checks the $q = 17$ class reduction or the $q = 19$ extension reduction.

F.2 Rank-three realization certificates

The ten-point bundle consists of `check_match10_rank_three.py`, its JSON certificate, and `check_match10_rank_three.sha256` in the paper directory. It is fixed at companion commit `ab3bcd6e`; the checksum file there records the full SHA-256 digests of the checker and JSON. The JSON fields `integral_lifts`, `algebraic_checks`, and `classical_char2_lex_basis` record respectively the rational unit relations and denominator primes, the saturated-ideal acceptance checks, and the characteristic-two triangular basis. The companion gives the exact replay command. Singular Gröbner bases and module lifts remain trusted executions. A second monomial order and a direct $\mathbb{F}_8$ incidence construction provide independent checks; the same record checks the regular hyperoval, the disjoint prescribed conic used in Corollary 4.6, and its nonvanishing gradient. Abstract two-class completeness remains the cited external theorem. The odd-characteristic computation saturates by $x_9(x_9 - 1)$, the product of the two necessary arc determinants displayed in the proof of Theorem A.3, and the checked rational module lifts have no denominator prime other than 2; hence their unit relations survive in every odd characteristic.

F.3 The $q = 16$ augmentation and leaf certificates

For the $q = 16$ exclusion, the separately versioned `finitegeom-q16-certificates` library contains the source generator, its frozen report, their SHA-256 digests, and the emitted Lean modules. The nine-line report is only a generation summary, not the certificate: the proof objects are the transition and leaf modules consumed by the package gate. Each certified transition carries an invertible matrix and checked source–target scalar equalities. The Lean checker separately verifies that these transitions cover every legal augmentation of the standard four-frame. At the leaves it checks either six independent quadratic evaluation conditions or an explicit linear combination forcing a containing quadratic to meet the arc. Thus the theorem trusts the Lean kernel and the stated classical inputs, but neither the generator's canonical labels nor an external claim of exhaustive execution.

The independent pattern checker recomputes every ordinary-uncovered locus from the frozen level-eight list and verifies the line–triangle pattern on exactly 2630 leaves. Its canonical JSON output records one six-point witness per leaf and the three exceptional kernels. This checker supplies the geometric compression used in the proof and checks the two displayed factorizations and the middle form's invertible standard-conic coordinate change on all 16^3 coordinate vectors. The independently shaped Lean inverse matrices remain a kernel-checked verification of the same 2630+3 partition. The field-uniform normalized line–triangle argument is separately kernel-checked in `QuadraticLineTriangleObstruction`; the terminal theorem's determinant hypothesis is the coordinate form of noncollinearity for the three off-line points.

G Finite-field witnesses

All coordinates below are relative to the standard conic

$$\mathcal{C} : \; XZ = Y^2.$$

For $q = 8$ and $q = 16$, an integer is the binary coefficient vector of the polynomial basis $1, \alpha, \ldots$, with $\alpha^3 + \alpha + 1 = 0$ and $\alpha^4 + \alpha + 1 = 0$, respectively. For $q = 9$, the integer $a_0 + 3a_1$ denotes $a_0 + a_1\alpha$, where $\alpha^2 + 1 = 0$. Coordinates for $q = 5, 11, 13, 17, 19$ are read modulo the prime. The checked witnesses are

q	A
5	$\{(1,0,3),(2,1,2),(3,3,4),(1,4,4)\}$
8	$\{(0,1,1),(0,1,2),(1,0,1),(1,0,2),(1,1,2),(1,1,6)\}$
9	$\{(1,0,4),(1,0,5),(1,1,0),(1,1,2),(1,2,3),(1,2,4)\}$
11	$\{(1,10,0),(1,9,1),(1,4,7),(1,8,5),(0,1,4),(1,1,7)\}$
13	$\{(4,0,5),(3,12,7),(2,3,10),(9,2,9),$ $(7,1,0),(6,0,2),(8,12,11),(7,11,0)\}$
16	$\{(1,5,14),(1,5,3),(1,9,2),(1,0,9),(1,13,13),$ $(1,6,10),(1,3,9),(0,1,11),(1,10,2)\}$
17	$\{(0,11,0),(0,8,1),(3,5,15),(3,7,16),(3,3,0),$ $(3,0,1),(3,7,2),(3,12,7),(3,16,14)\}$
19	$\{(0,1,8),(1,1,0),(1,3,14),(1,9,0),(1,9,12),$ $(1,11,11),(1,13,9),(1,16,1),(1,16,12),(1,18,7)\}$

The files `lean/RelativeConicArcs/ExampleChecks/Q5.lean` through `Q16.lean` check disjointness from $\mathcal{C}$, the arc condition, and relative coverage. The independent verifier `verify_relative_conic_arcs.py` regenerates the same coverage data; its digest and frozen output are recorded in the trust manifest. The public supplement gate `RelativeConicArcs/Gates/ArcsCompleteOutsideConicAdditions.lean` checks the three prime-field witnesses just added, including ordinary completeness of the $q = 19$ arc.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author's direction, the systems assisted throughout with proof exploration, literature research, symbolic and finite computation, code and formal-proof development, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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